# Lab Report

## Spectral Analysis, Adaptive Noise Cancelling, and Blind Source Separation for ECG Signals


## Submitted by ABBASI Muhammad Hasham Waseem


Pregnancy abdominal/thoracic ECG and atrial fibrillation ECG analyses performed in MATLAB Online
This report consolidates all figures, tables, numerical results, and interpretive discussion requested in the lab assignment.

MATLAB code and files can be found in the following repository: https://github.com/mhwa06/Biomedical-Signal-Processing-Lab-1.git

| | |
|---|---|
| **Report scope** | Assignment Parts 1.1-3.4 |
| **Datasets used** | Pregnancy ECG: 8 leads × 5000 samples at 500 Hz; AF dataset: 75 patients, 12 leads × 15000 samples |
| **Labeled AF records** | 63 of 75 patients had recurrence labels available |
| **Important implementation note** | AF sampling frequency was set to 250 Hz in the analysis configuration because the uploaded AF files did not explicitly store fs |

06 April 2026

# 1. Introduction and objectives

This lab assignment evaluated three families of ECG signal-processing techniques on synthetic and real data: (i) spectral analysis and filtering, (ii) multi-reference adaptive noise cancelling (MRANC), and (iii) blind source separation (BSS) using principal component analysis (PCA) and independent component analysis (ICA). The real-data tasks focused on two clinically relevant problems: extraction of fetal ECG (FECG) from mixed abdominal recordings of a pregnant woman, and extraction/characterization of atrial activity (AA) in atrial fibrillation (AF) ECG recordings.

The professor explicitly requested that the lab report should not merely present figures and tables, but should interpret and discuss them. Accordingly, each section below combines the numerical outputs produced in MATLAB Online with comments explaining what the figures mean, what conclusions can be drawn, and where the limitations of each method appear.

# 2. Data, signal models, and analysis settings

Pregnancy ECG record: the uploaded file contained an 8-lead recording with 5000 samples per lead and a sampling frequency of 500 Hz. The first five leads are abdominal recordings (Abd1-Abd5), which contain mixtures of maternal ECG (MECG), fetal ECG (FECG), and noise. The last three leads are thoracic recordings (Thor1-Thor3), dominated by maternal ECG and used as references in the ANC experiments. This matches the classic benchmark data structure used for comparing blind separation with adaptive cancellation in fetal ECG extraction.

AF dataset: the provided files combined into a dataset of 75 patients, with 15000 samples × 12 leads per patient. Of these, 63 patients had recurrence labels and could therefore be included in the recurrence-prediction experiments. During code execution, the AF sampling frequency was set to 250 Hz in the configuration file because the uploaded MAT files did not explicitly contain fs. This is an honest implementation detail and should be updated if the professor's original dataset documentation specifies a different sampling rate.

Spectral parameters used for AF analysis:

- Dominant frequency (DF): frequency location of the maximum PSD peak within the AF band [3, 9] Hz.
- Spectral concentration (SC): percentage of signal power concentrated in a narrow band around DF. In this report, the classical band [0.82·DF, 1.17·DF] was used around the dominant frequency.
- PSD estimator: Welch method, chosen because it is the conventional practical estimator for ECG spectral characterization and provides robust average spectral estimates.

For the BSS sections, PCA was performed with the professor's supplied function bss_pca.m, which uses singular value decomposition on the observation matrix X and returns an estimated mixing matrix H and normalized source matrix S such that $X \approx H \cdot S$. For ICA, an ICA-based separation stage was used to obtain independent components and their projections back to the observed leads.

# 3. Spectral analysis and filtering

## 3.1 Abdominal ECG during pregnancy (Assignment 1.1)

The first task was to inspect the 8 recorded leads and determine which abdominal channel was most strongly contaminated. Visual inspection clearly showed that the fourth abdominal lead (Abd4) was noisier than the others. Its waveform presented slower baseline drift and visibly stronger random fluctuations superimposed on the ECG morphology.

*Figure 1. Eight-lead pregnancy ECG record. The first five leads are abdominal and the last three leads are thoracic. Abd4 is visibly noisier than the other abdominal leads.*

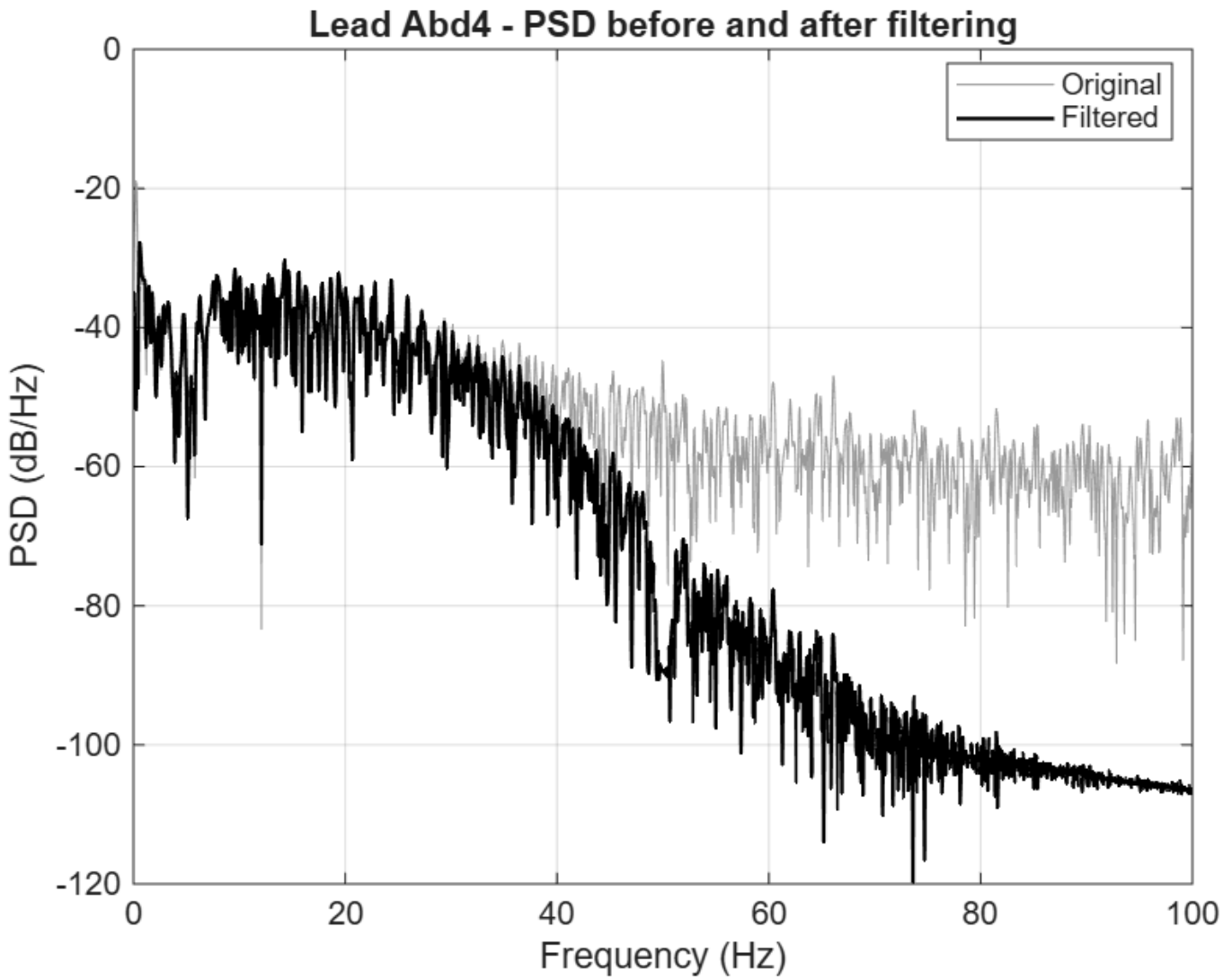


*Figure 2. Time-domain view of the noisy abdominal lead Abd4 before and after filtering. The filtered trace is cleaner, although a short start-up transient remains near the beginning of the record.*

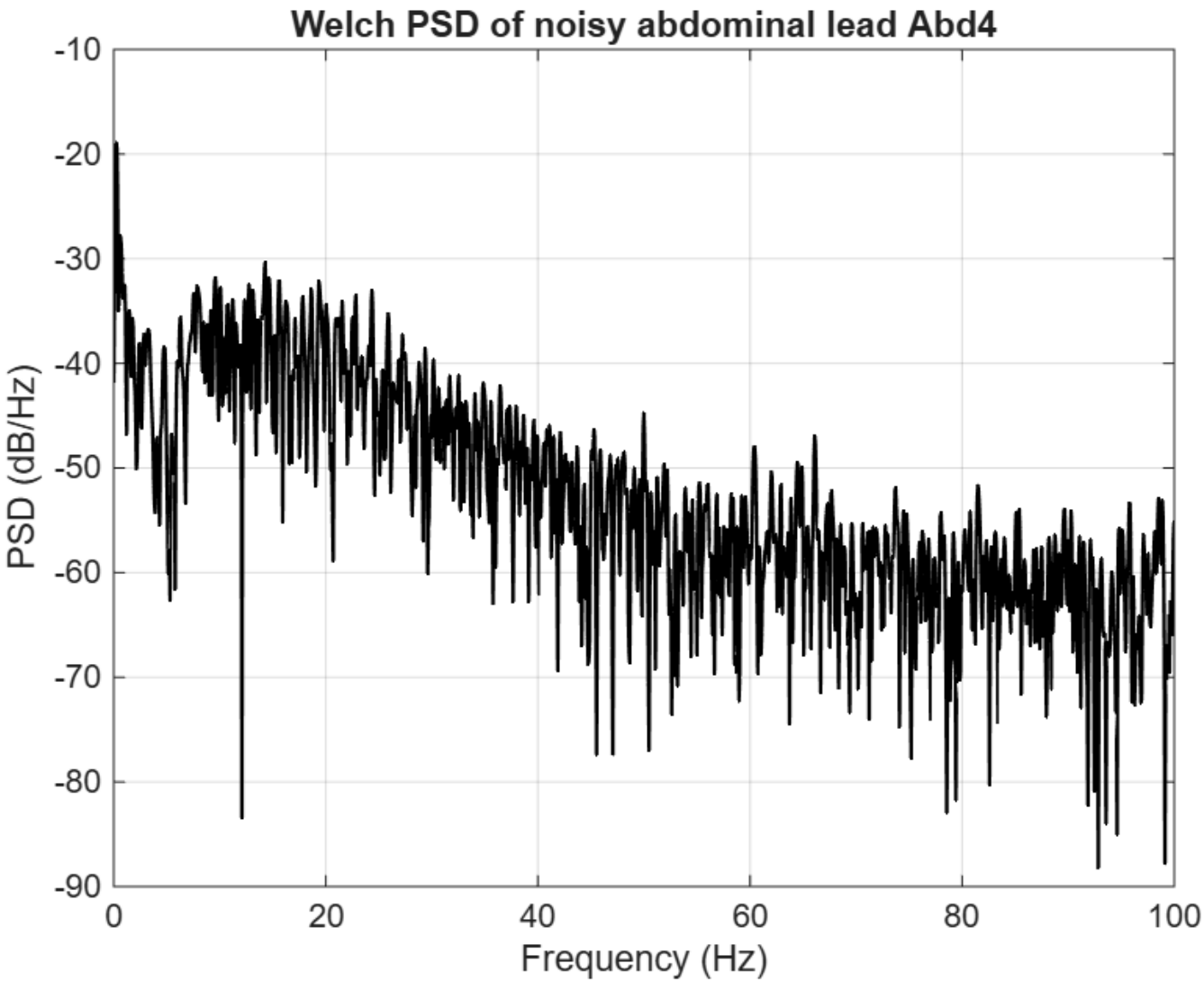


***Figure 3. Welch PSD estimate of the noisy abdominal lead Abd4. Spectral inspection reveals strong low-frequency content, a narrow interference near mains frequency, and broad high-frequency noise.***

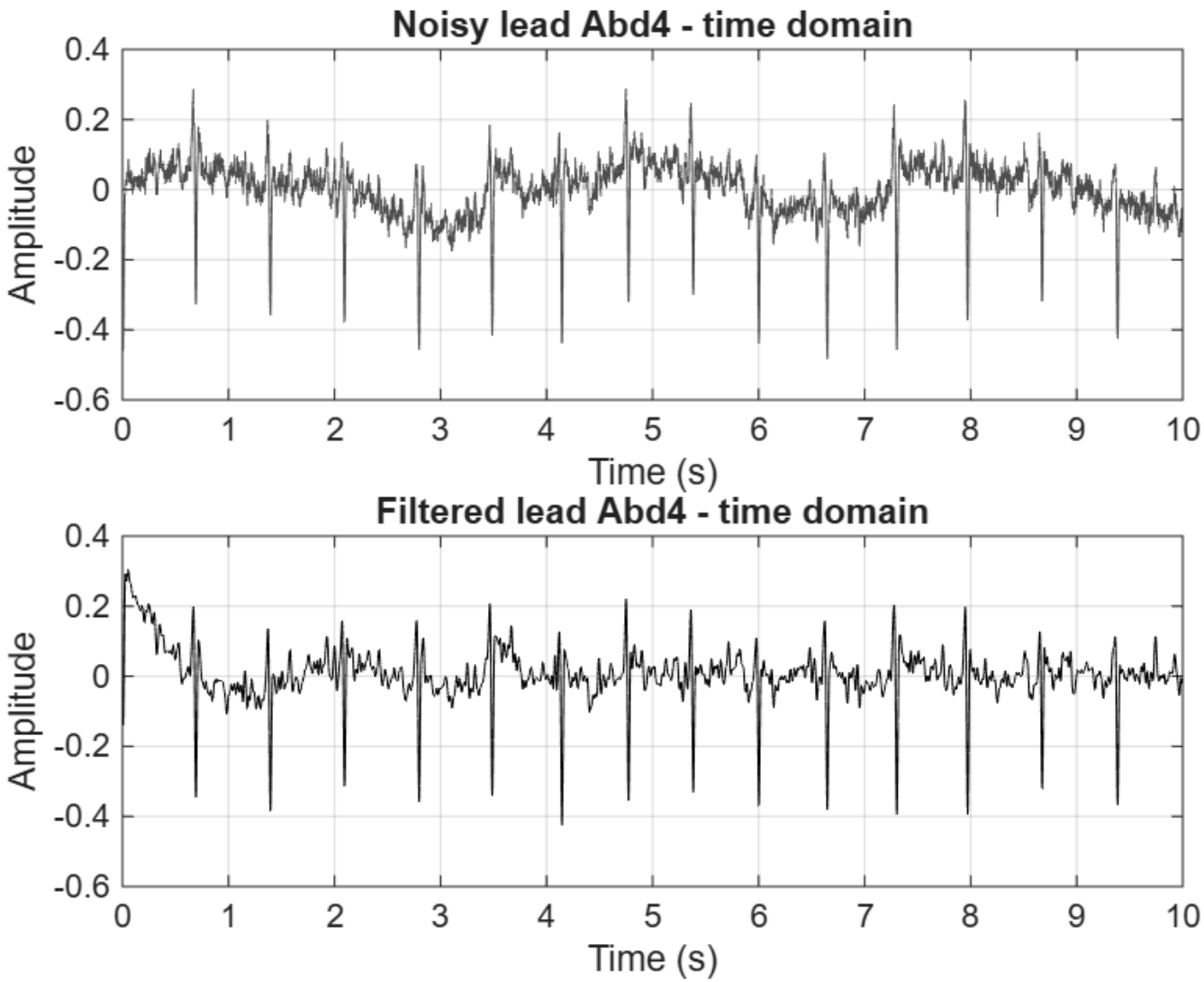


***Figure 4. PSD of Abd4 before and after filtering. The low-frequency baseline wander, narrowband 50 Hz interference, and broad high-frequency noise are markedly attenuated after filtering.***

**Table 1. Noise and interference identified in noisy abdominal lead Abd4**

| Observed contamination | Approximate frequency region | Interpretation | Filtering action used |
|---|---|---|---|
| Very-low-frequency drift | Below about 0.5 Hz | Baseline wander from motion/respiration and slow electrode variations | High-pass filtering (0.5 Hz) |
| Narrowband interference | Around 50 Hz | Power-line contamination | 50 Hz notch filtering |
| Broadband high-frequency noise | Above about 40 Hz | Muscle/electronic/high-frequency noise | Low-pass filtering (40 Hz) |

The PSD of Abd4 confirmed three disturbances that are textbook ECG contaminants: (1) baseline wander below approximately 0.5 Hz, (2) a mains-interference component around 50 Hz, and (3) broadband noise above roughly 40 Hz. A cascade of high-pass, notch, and low-pass filters was therefore appropriate.

The post-filter PSD shows a strong attenuation of unwanted content while preserving the spectral region where the diagnostically relevant ECG energy lies. In the time domain, the filtered signal is visibly cleaner and the QRS complexes remain identifiable. The short transient at the beginning of the filtered record is an edge effect due to filtering and should not be interpreted as physiology.

Hence, the filtering stage achieved its main objective: it reduced baseline drift, suppressed narrowband power-line interference, and attenuated high-frequency noise while keeping the ECG morphology interpretable. For this dataset, the filter design was therefore effective and well matched to the PSD inspection.

## 3.2 Atrial fibrillation spectral analysis (Assignment 1.2)

The AF task had two goals. First, DF and SC had to be computed on the recorded ECGs (Xva) and on the provided atrial estimates (Xa). Second, these features had to be tested as predictors of AF recurrence using 5-fold cross-validation. Lead V1 was emphasized because atrial activity is typically most visible there, and an additional experiment averaged the spectral parameters over the six precordial leads V1-V6.

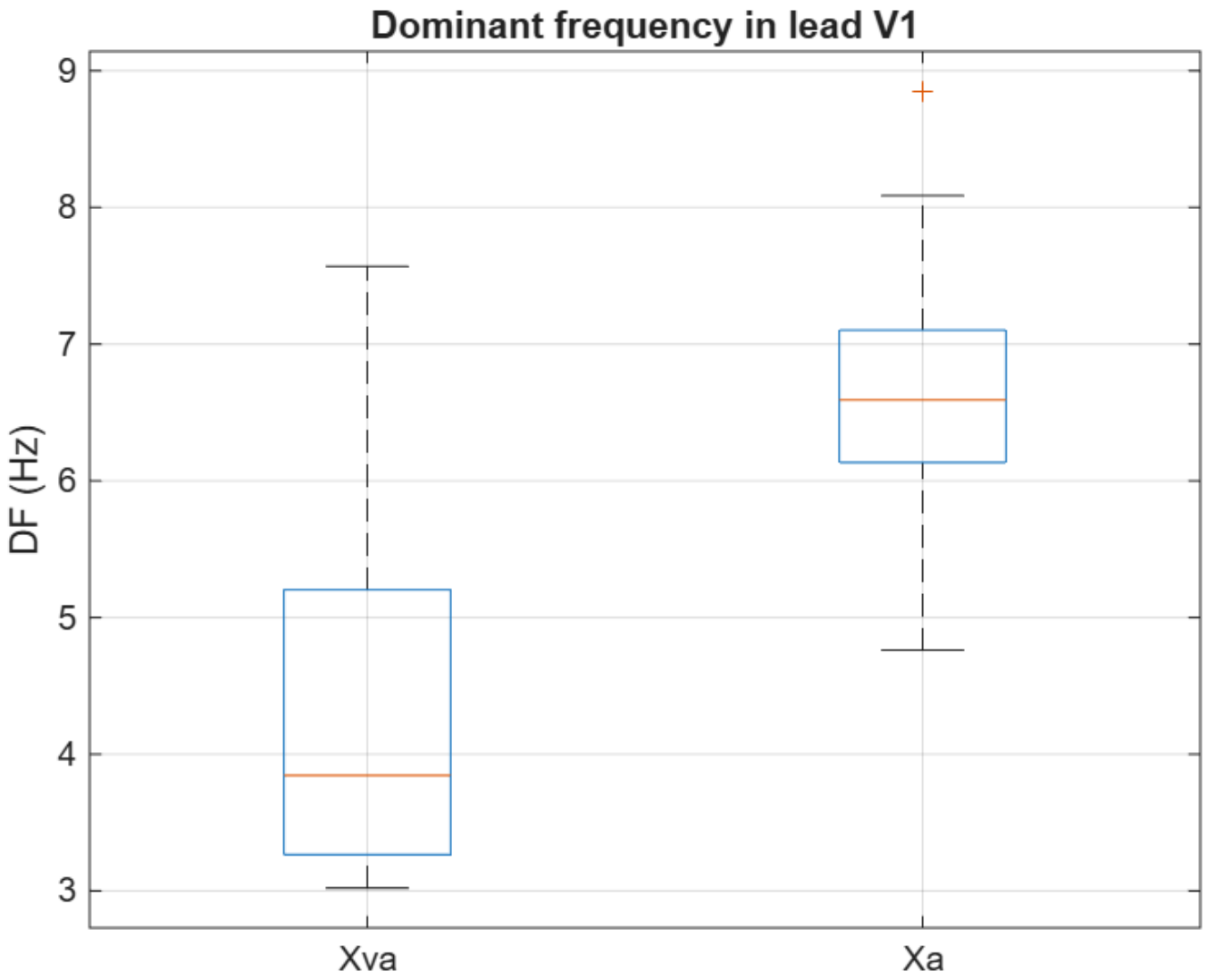


***Figure 5. Box-and-whisker plots of dominant frequency (DF) in lead V1 for the recorded ECG (Xva) and the provided atrial estimate (Xa). Xa shifts the dominant spectral energy more clearly into the AF band.***

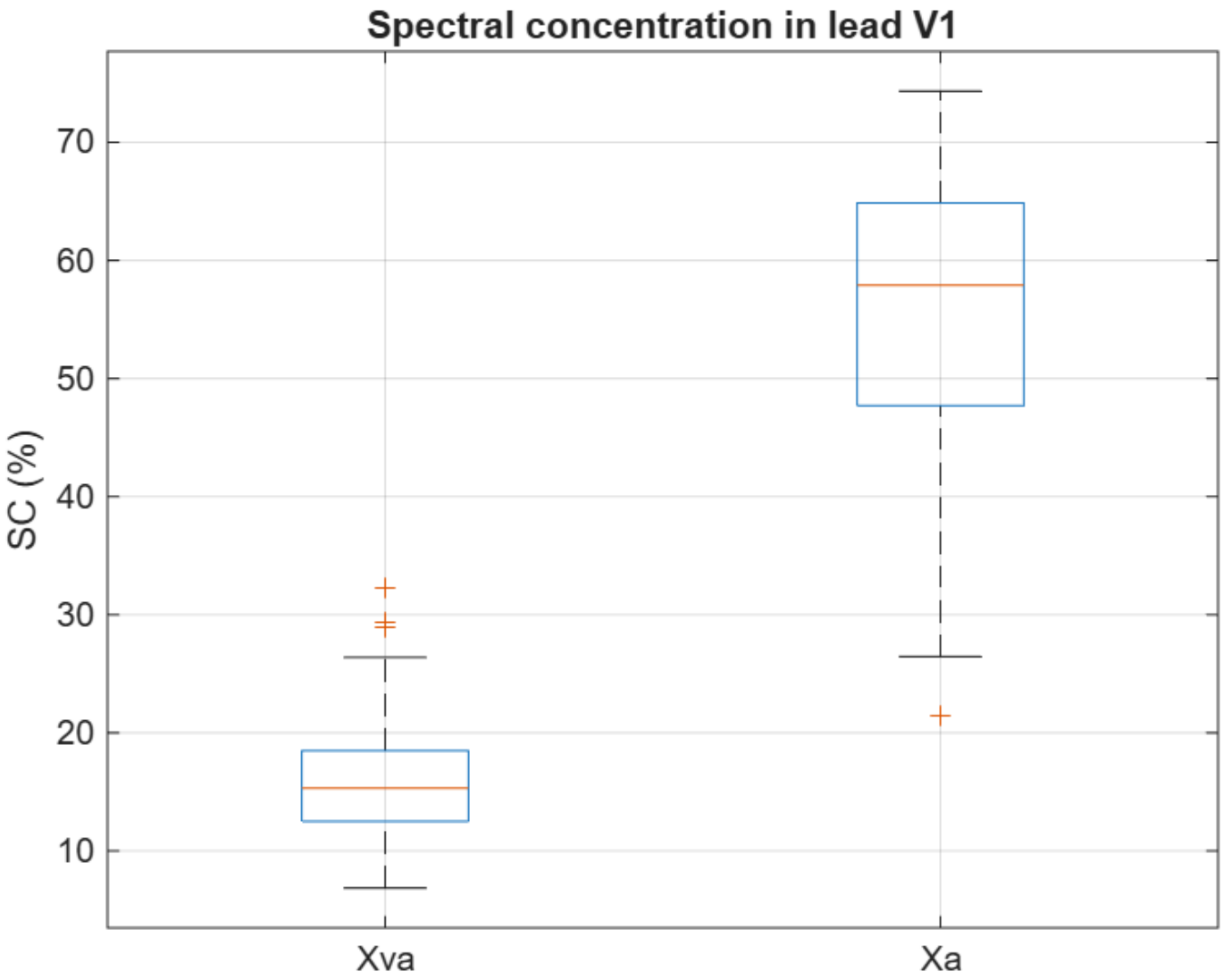


***Figure 6. Box-and-whisker plots of spectral concentration (SC) in lead V1 for Xva and Xa. The provided atrial estimate Xa shows much higher spectral concentration than the raw recording.***

### Table 2. Median DF and SC values over labeled AF patients

| Feature set | Median DF (Hz) | Median SC (%) | Interpretation |
|---|---|---|---|
| V1 of Xva | 3.723 | 15.309 | Raw V1 is heavily contaminated; atrial rhythm is less cleanly represented |
| V1 of Xa | 6.592 | 58.257 | Provided atrial estimate is much cleaner and more periodic |
| Average of V1:V6 in Xva | 4.450 | 14.515 | Averaging slightly shifts DF upward but does not clean the signal |
| Average of V1:V6 in Xa | 6.454 | 44.887 | Atrial estimate remains organized but less concentrated than V1 alone |

### Table 3. Five-fold LDA classification of AF recurrence using DF and SC

| Feature set | Accuracy | Sensitivity | Specificity | Precision | F1-score | Balanced accuracy |
|---|---|---|---|---|---|---|
| V1 features from Xva | 0.492 | 0.625 | 0.355 | 0.500 | 0.556 | 0.490 |
| V1 features from Xa | 0.492 | 0.469 | 0.516 | 0.500 | 0.484 | 0.492 |
| Average V1:V6 from Xva | 0.556 | 0.594 | 0.516 | 0.559 | 0.576 | 0.555 |
| Average V1:V6 from Xa | 0.476 | 0.562 | 0.387 | 0.486 | 0.522 | 0.475 |

The boxplots and medians show a very consistent result: Xa is spectrally much cleaner than Xva. Its DF is centered in the AF band and its SC is far higher, which means the atrial estimate has a more organized spectrum and less residual ventricular contamination. In contrast, Xva still contains QRST and other interfering components, so the dominant spectral content is shifted downward and the energy is more diffuse.

Lead V1 remains particularly informative. When the precordial leads are averaged, the raw recordings gain only a slight increase in DF and no meaningful improvement in SC. For the atrial estimates Xa, averaging V1:V6 preserves a reasonable DF but decreases SC compared with V1 alone, indicating that the strongest atrial signature was already present in V1 and averaging diluted it.

The recurrence-classification results are modest. The best balanced accuracy was 0.555 when the averaged V1:V6 features were extracted from Xva. All other cases hovered close to chance level. Therefore, in this dataset DF and SC are valuable descriptive biomarkers for signal organization, but they are not strong stand-alone predictors of AF recurrence.

In summary, the spectral parameters clearly distinguish raw ECG recordings from cleaner atrial estimates, but they do not provide sufficiently strong discrimination by themselves to predict recurrence reliably in the present population.

# 4. Adaptive noise cancelling

## 4.1 Synthetic signals (Assignment 2.1)

The synthetic ANC experiment evaluated the influence of the LMS step size μ and the filter order M on multi-reference ANC performance. The primary input consisted of the desired signal plus interference, while three reference channels carried interference information for cancellation.

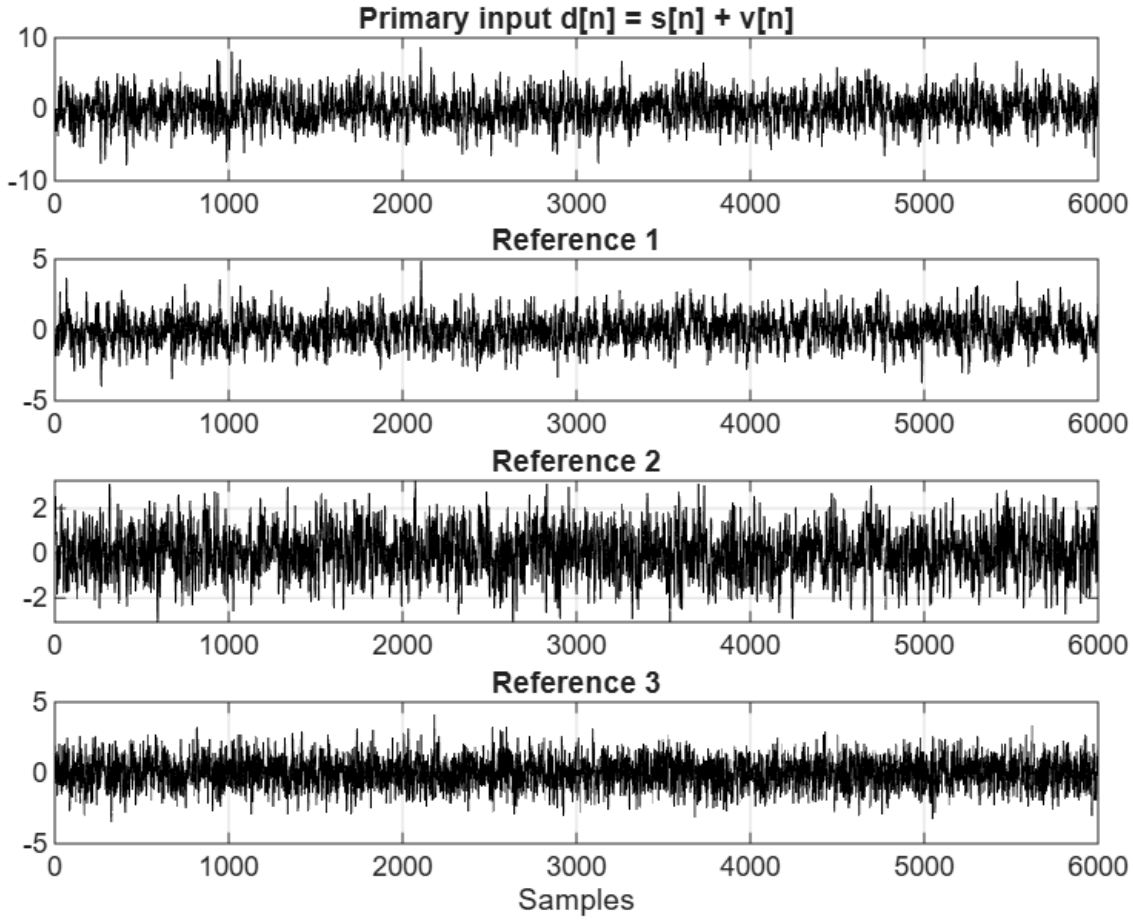


*Figure 7. Synthetic ANC setup: primary input d[n] = s[n] + v[n] and three interference reference signals.*

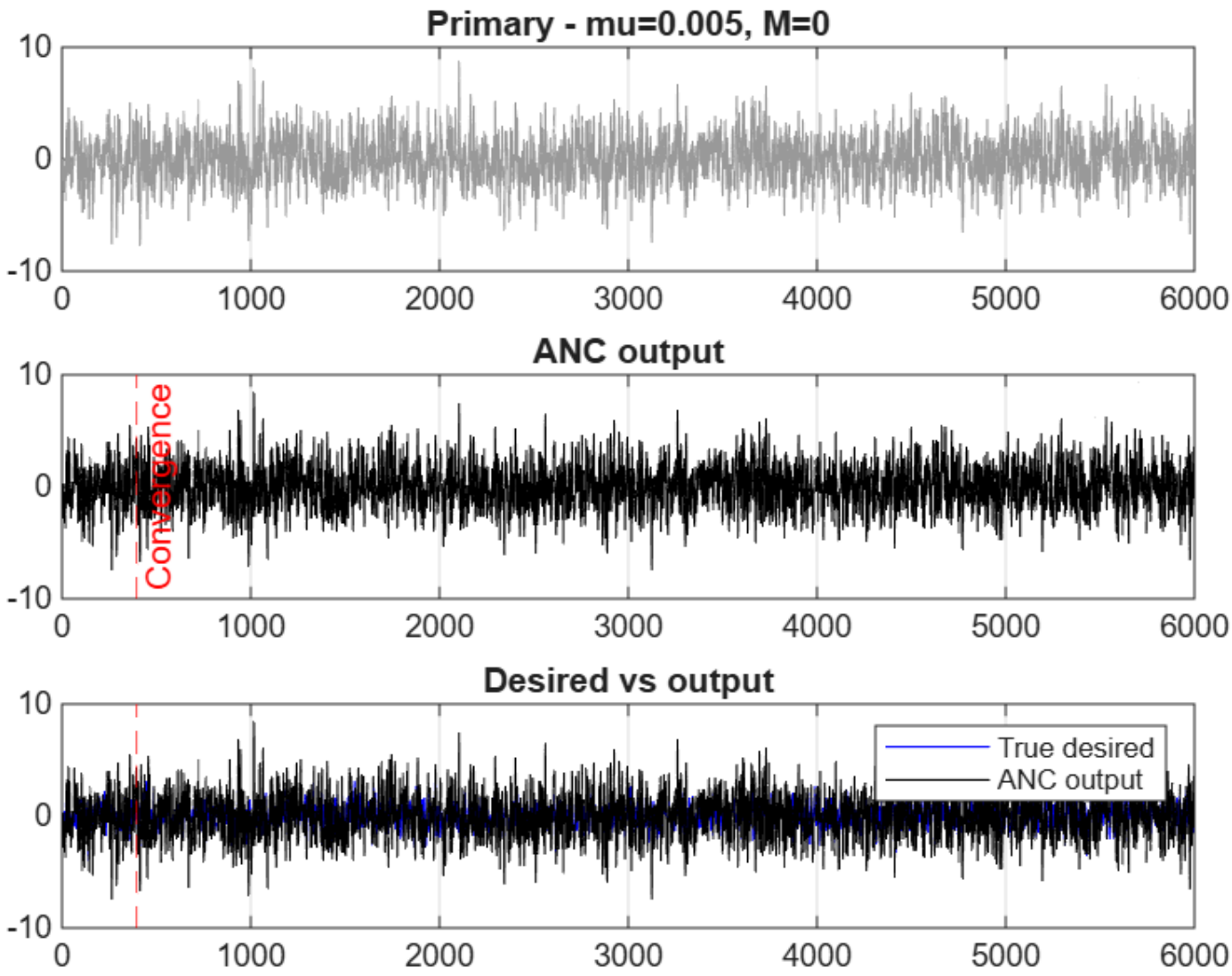


*Figure 8. ANC output for μ = 0.005 and M = 0. The convergence point is marked, and the output only moderately improves the desired-signal recovery.*

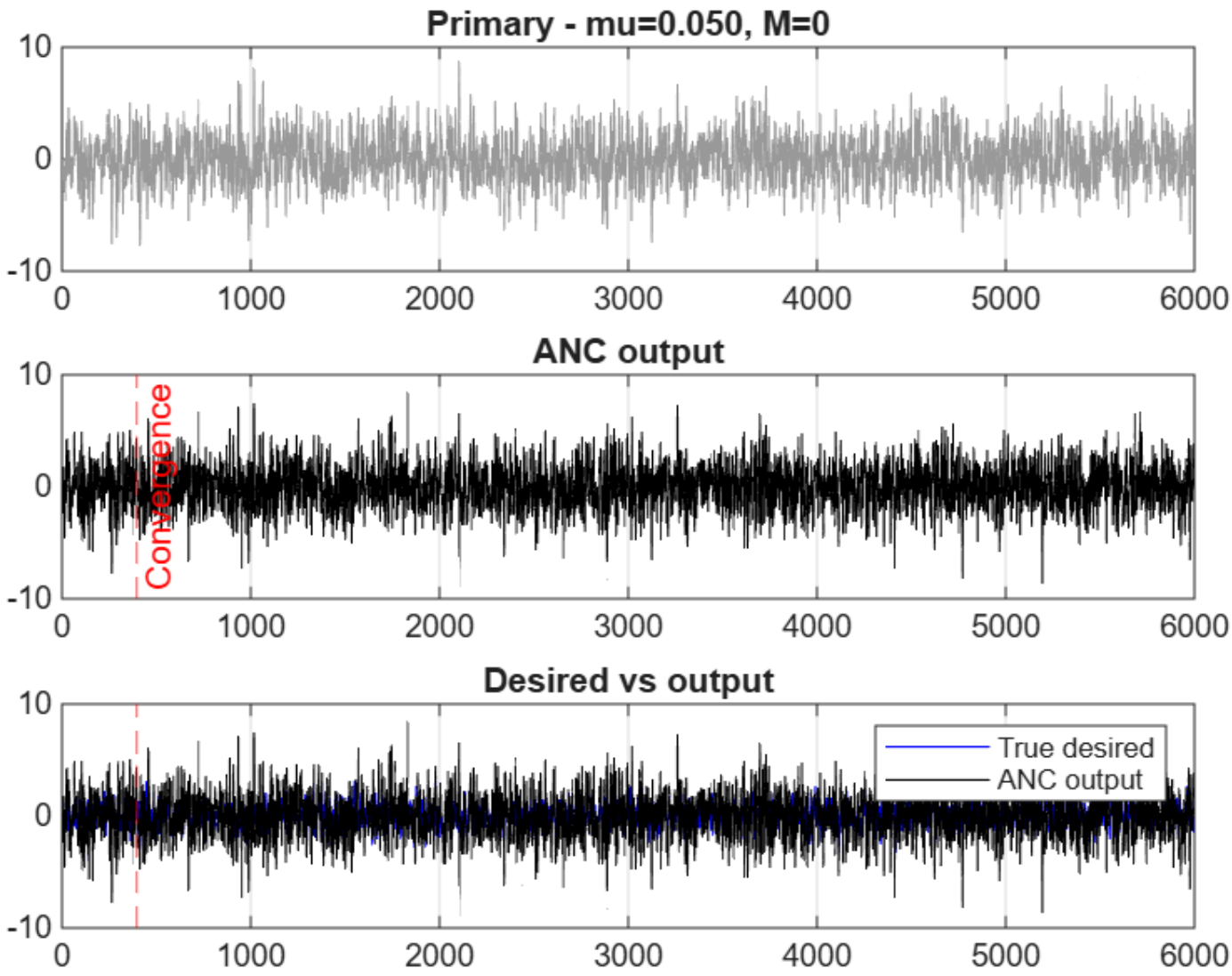


*Figure 9. ANC output for μ = 0.050 and M = 0. The large step size degrades steady-state quality even though adaptation is fast.*

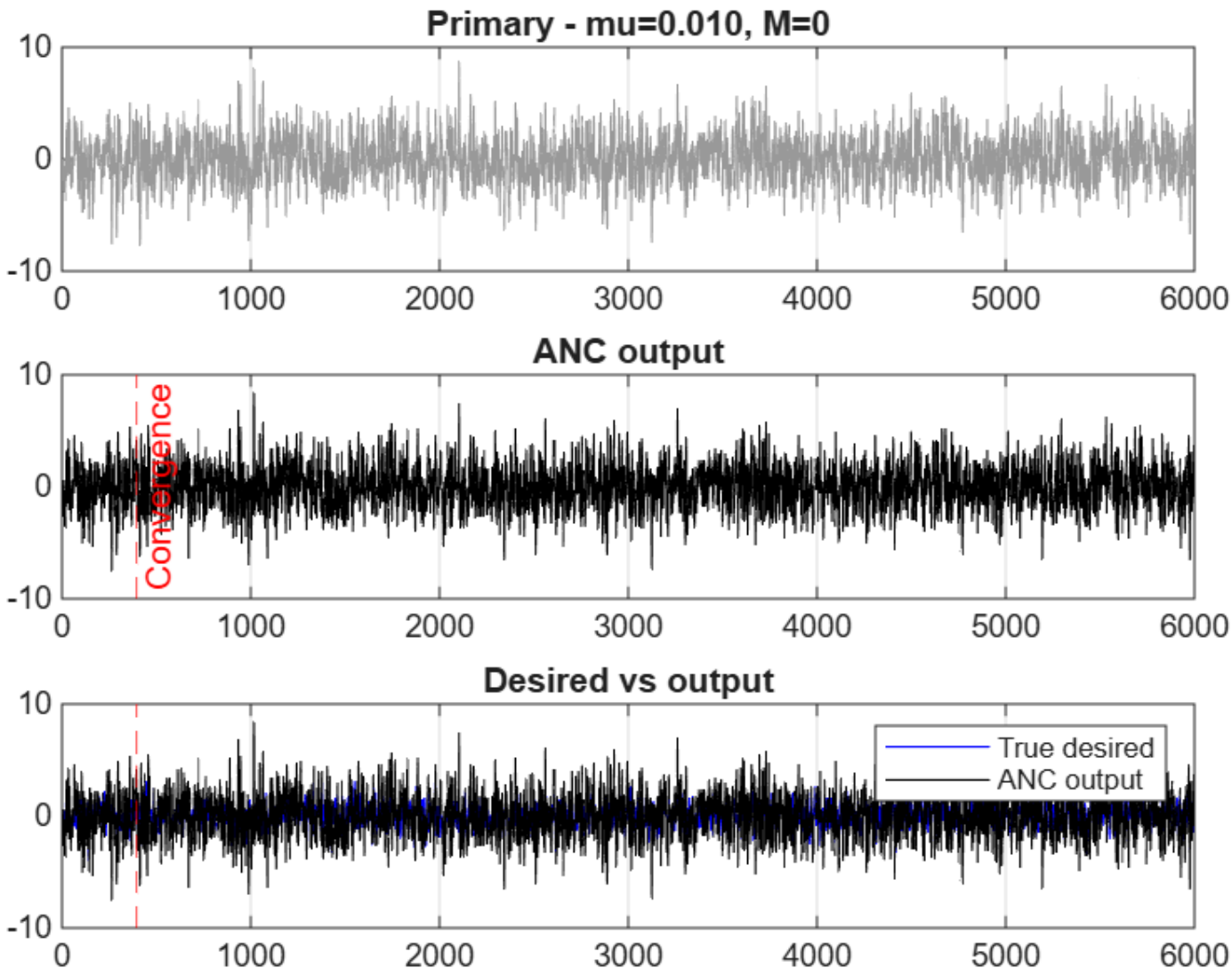


*Figure 10. ANC output for μ = 0.010 and M = 0. This intermediate step size remains stable but only slightly improves the output SIR.*

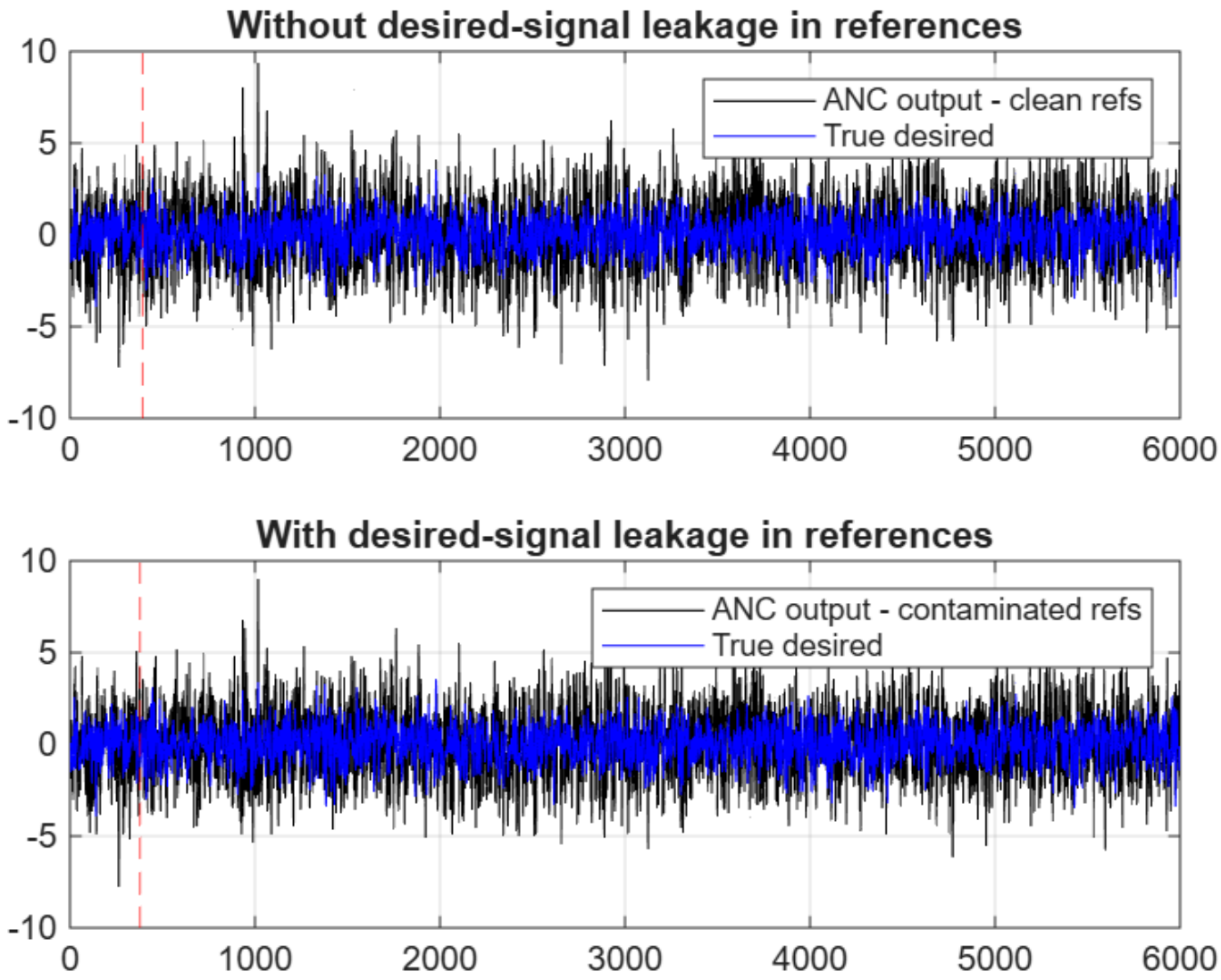


***Figure 11. Effect of desired-signal leakage into the reference channels. Leakage causes partial cancellation of the desired signal itself and visibly degrades the recovered output.***

### Table 4. Synthetic MRANC parameter study

| Test case | Input SIR (dB) | Output SIR (dB) | SIR improvement (dB) | Convergence sample |
|---|---|---|---|---|
| μ = 0.005, M = 0 | -4.9233 | -4.5060 | +0.4173 | 395 |
| μ = 0.050, M = 0 | -4.9195 | -5.0535 | -0.1340 | 396 |
| μ = 0.010, M = 0 | -4.9233 | -4.5625 | +0.3609 | 395 |
| μ = 0.010, M = 5 | -4.9308 | -5.0605 | -0.1297 | 393 |
| μ = 0.010, M = 50 | -4.9198 | NaN | NaN | 1200 |

### Table 5. Synthetic MRANC leakage experiment

| Reference condition | Output SIR (dB) | Interpretation |
|---|---|---|
| Clean references | -5.061 | Baseline ANC performance with interference-only references |
| Leaky references | -5.246 | Lower output SIR because the filter starts cancelling part of the desired signal |

The parameter study shows that the step size must be chosen carefully. The smaller values μ = 0.005 and μ = 0.010 gave small positive SIR improvements, whereas μ = 0.050 degraded steady-state performance. This is consistent with LMS theory: a larger step size can speed up adaptation, but if it is too large it increases misadjustment and can destabilize the estimate.

The filter order study further showed that a larger M is not automatically beneficial. In this synthetic configuration, M = 0 gave the best result, M = 5 worsened the output, and M = 50 effectively failed. This indicates that adding many adaptive taps is only worthwhile when the interference path is genuinely time-dispersive; otherwise it increases model complexity and makes convergence more difficult.

Leakage of the desired signal into the references degraded performance, as expected. Once the reference channels contain a component of the signal of interest, ANC no longer performs pure interference cancellation; it also removes part of the desired signal. This causes distortion and lowers output SIR. Therefore, the reference channels must be as clean as possible in any practical ANC system.

## 4.2 Fetal ECG extraction with MRANC (Assignment 2.2)

In the pregnancy ECG, the abdominal leads were treated as primary mixtures and the thoracic leads as reference channels dominated by MECG. The objective was to suppress the maternal component and reveal fetal activity more clearly, under the assumption of no time dispersion (M = 0).

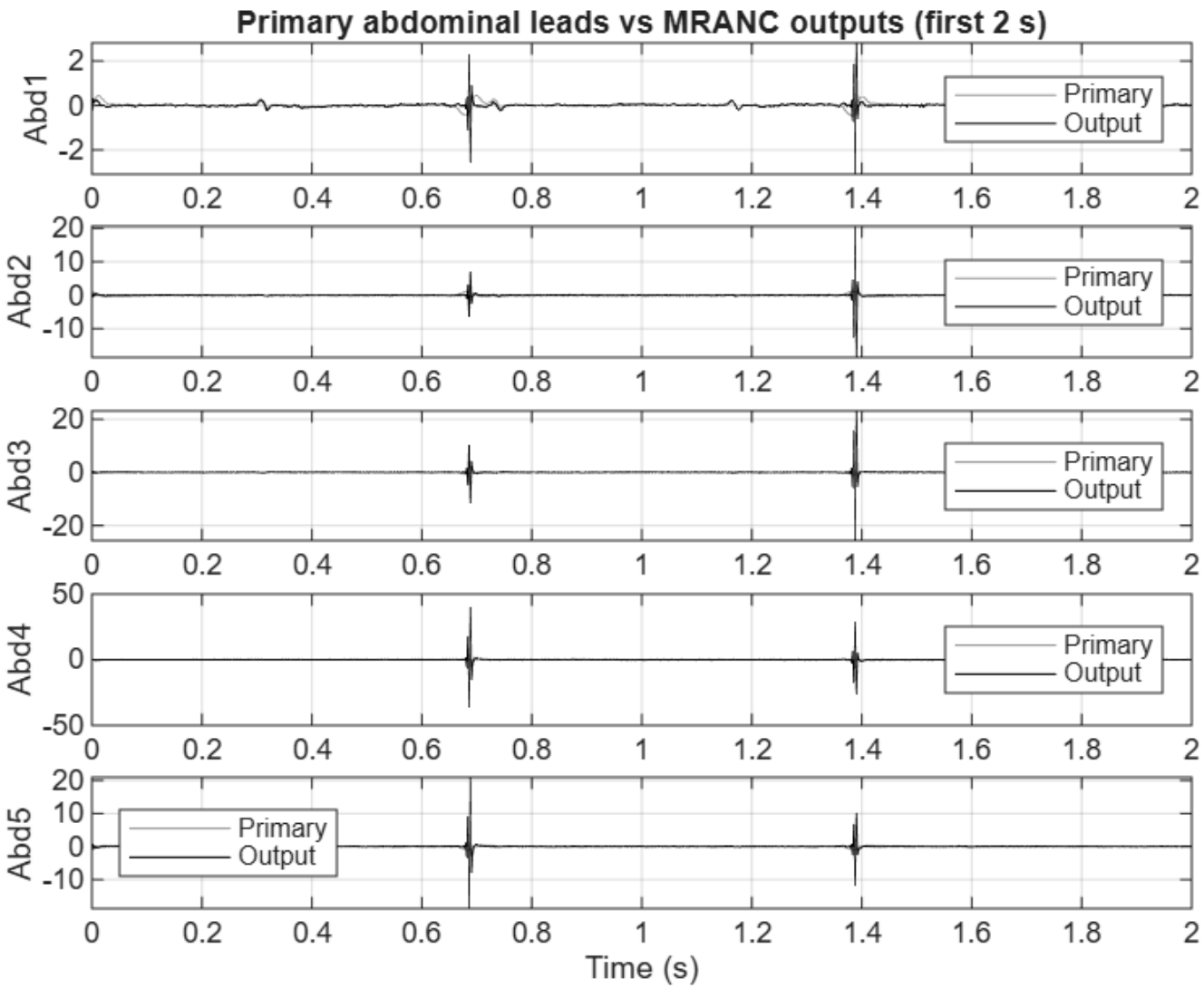


***Figure 12. First 2 seconds of each abdominal lead before and after MRANC. The output differs from the primary input, indicating adaptation, but the largest maternal complexes remain visible.***

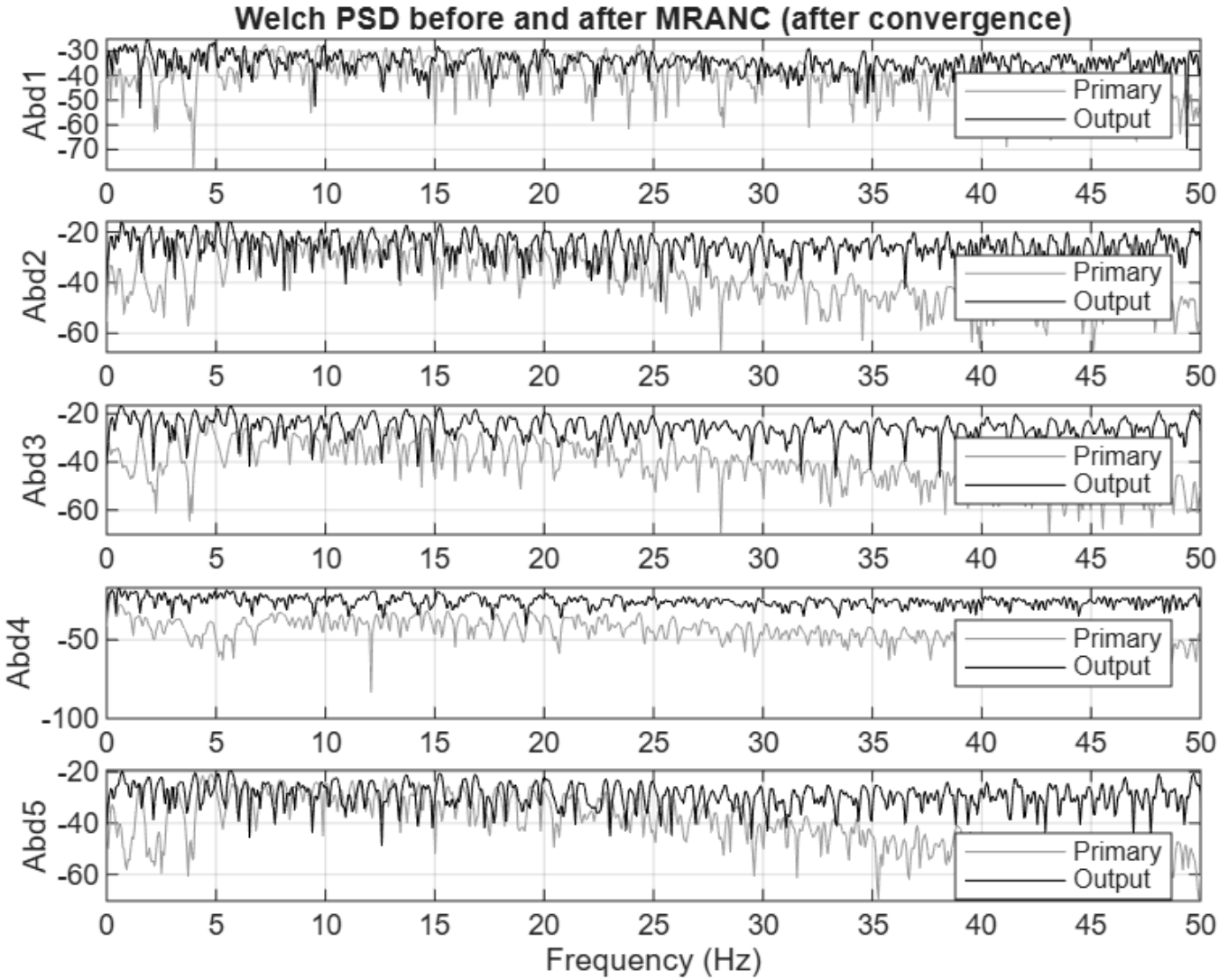


***Figure 13. Welch PSD comparison between primary abdominal leads and MRANC outputs (after convergence). The output spectra are modified, but substantial residual ECG energy remains.***

The MRANC output changes the abdominal signals and therefore does achieve some maternal cancellation. However, the dominant recurring complexes are still clearly visible in the output traces, which means the maternal ECG was only partially removed.

The PSDs after convergence are also altered relative to the primary inputs, confirming that the adaptive filter removed part of the common interference. Nevertheless, the remaining energy distribution shows that the maternal contribution was not fully cancelled. In practical terms, MRANC improved the signal but did not produce a clean, isolated fetal ECG in these recordings.

This partial performance is understandable. The maternal ECG seen in the abdomen is not an exact scaled copy of the thoracic ECG, and the simplified instantaneous-mixture assumption (M = 0) can limit cancellation quality. Thus, this experiment demonstrates the usefulness of MRANC but also its sensitivity to mismatch between the primary and reference pathways.

## 4.3 Atrial activity extraction for a single AF patient with MRANC (Assignment 2.3)

For the first AF patient, the six precordial leads V1-V6 were inspected and lead V1 was selected as primary because atrial activity is usually most visible there. The reference inputs were the neighboring precordial leads V2-V6, and the goal was to suppress QRST activity and retain atrial information.

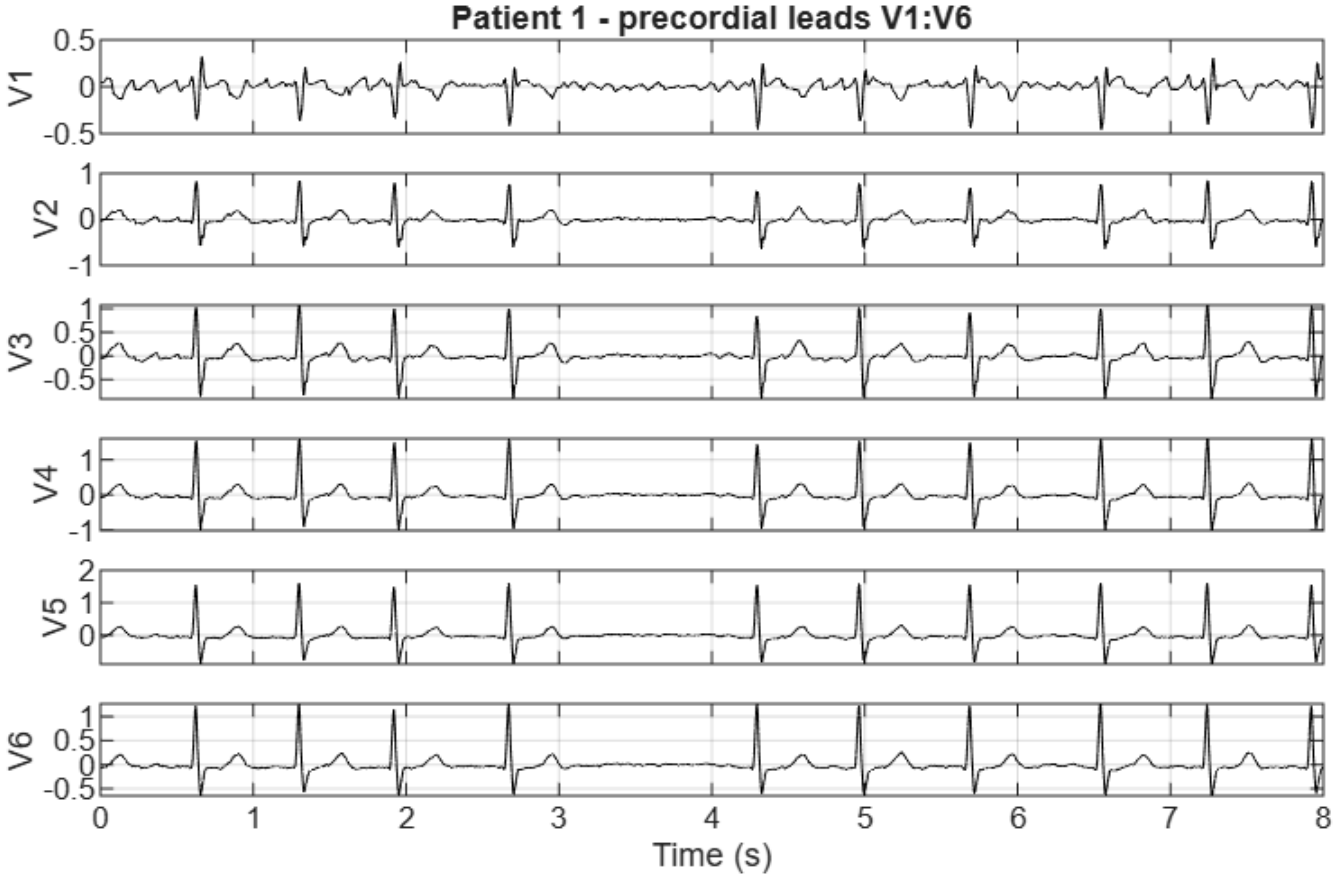

*Figure 14. Precordial leads V1-V6 for patient 1. Lead V1 is a suitable primary lead because atrial activity is comparatively more visible there.*

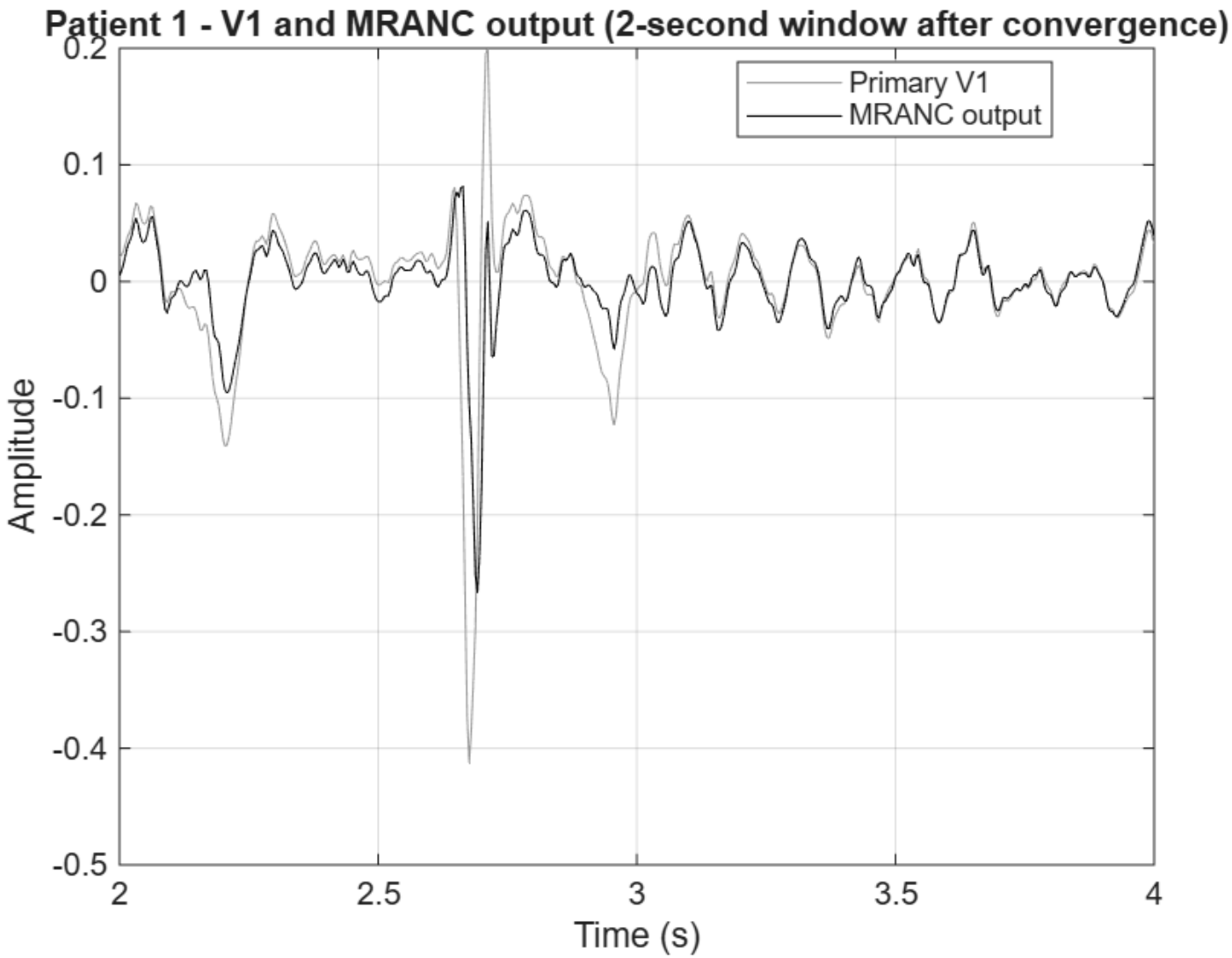


*Figure 15. Two-second window after convergence comparing patient 1 lead V1 with the MRANC output. Ventricular activity is attenuated but not eliminated.*

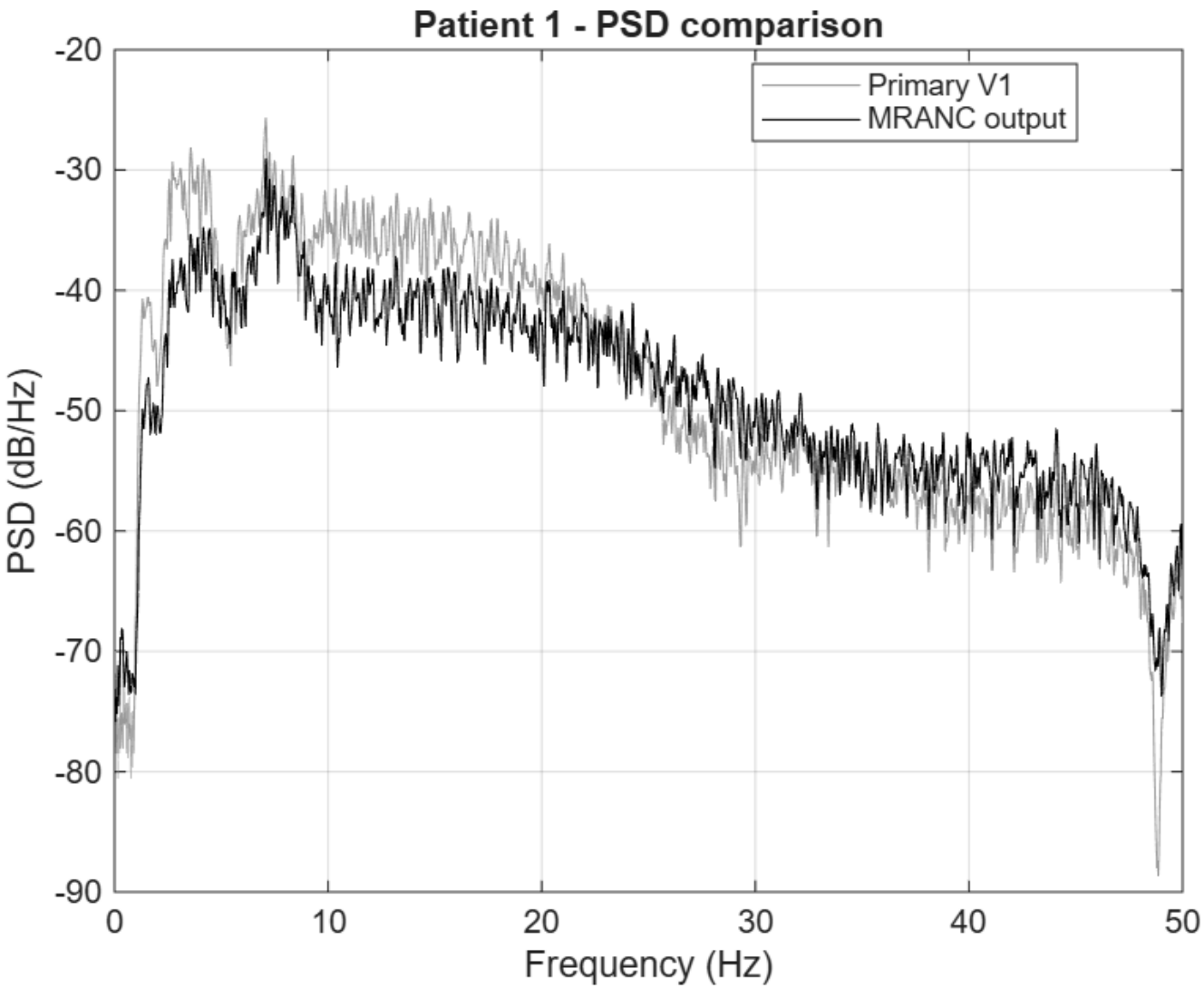


***Figure 16. PSD comparison between patient 1 lead V1 and its MRANC output. The output spectrum differs from the input, but the change is moderate rather than dramatic.***

**Table 6. Patient 1 atrial spectral parameters after MRANC and STC**

| Method | DF (Hz) | SC (%) | Interpretation |
|---|---|---|---|
| MRANC output | 7.080 | 29.893 | Recovers the main AF rhythm but with moderate spectral purity |
| STC atrial estimate (Xa V1) | 7.080 | 64.292 | Same dominant rhythm, but much cleaner and more organized |

The time-domain comparison shows partial ventricular suppression: the MRANC output is closer to a smooth atrial waveform than the raw V1 trace, but residual QRST complexes remain. In the frequency domain, the MRANC output has the same DF as the STC reference estimate (7.080 Hz), which means it preserves the principal atrial rhythm.

The key difference lies in SC. The MRANC output has SC = 29.893%, whereas the STC estimate has SC = 64.292%. Since a higher SC implies a cleaner, more organized narrowband atrial signal, STC clearly gives the better atrial extraction for this patient. MRANC preserves the correct rhythm but leaves substantially more residual contamination.

## 4.4 Atrial activity extraction over the full AF database with MRANC (Assignment 2.4)

The MRANC procedure was then repeated across the full database. Instead of plotting all patients, performance was summarized using DF and SC distributions over the 63 labeled patients and compared against the provided STC atrial estimates Xa.

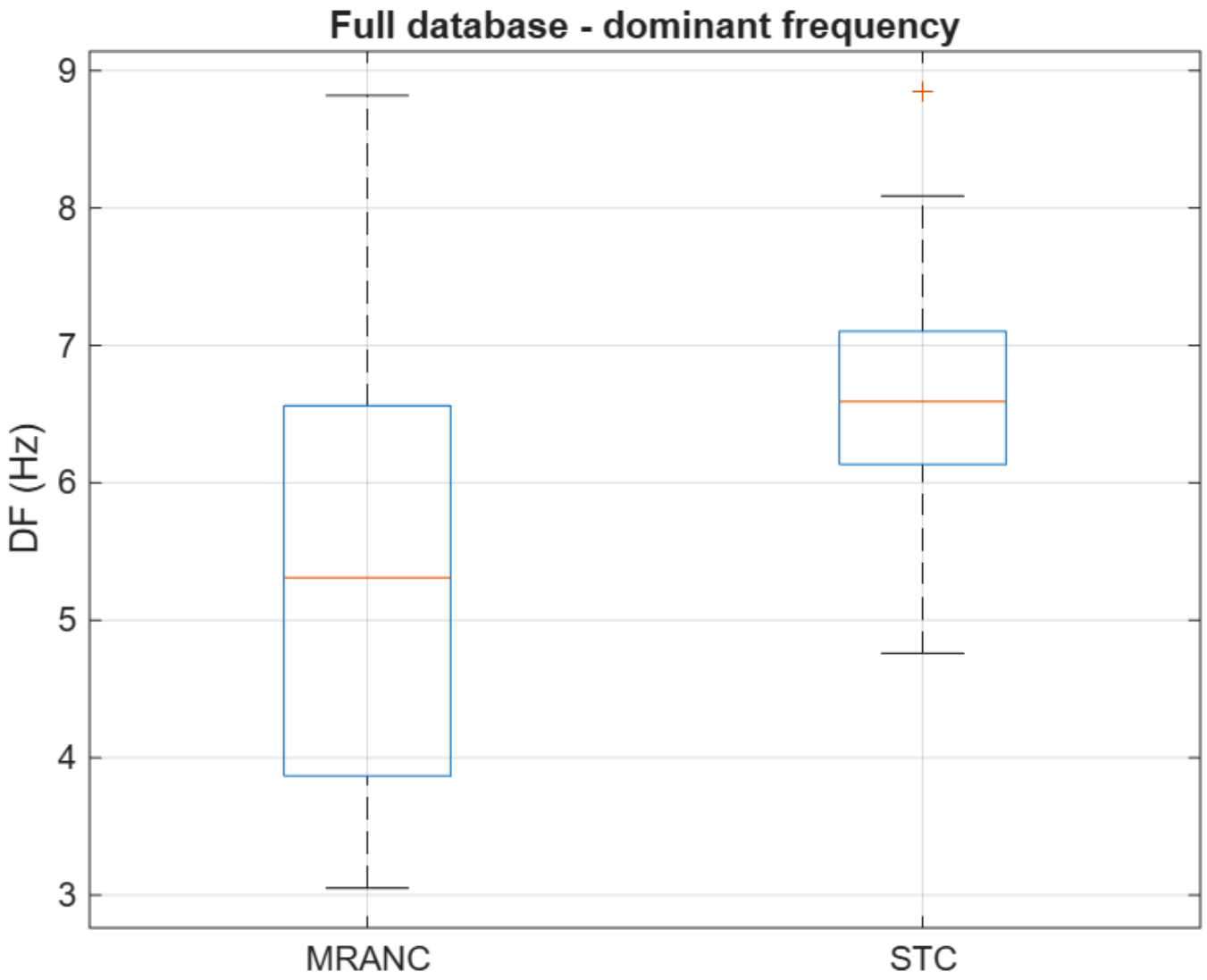


***Figure 17. Full-database comparison of dominant frequency for MRANC and STC atrial estimates. MRANC remains in the AF band but tends to give lower and more dispersed DF values.***

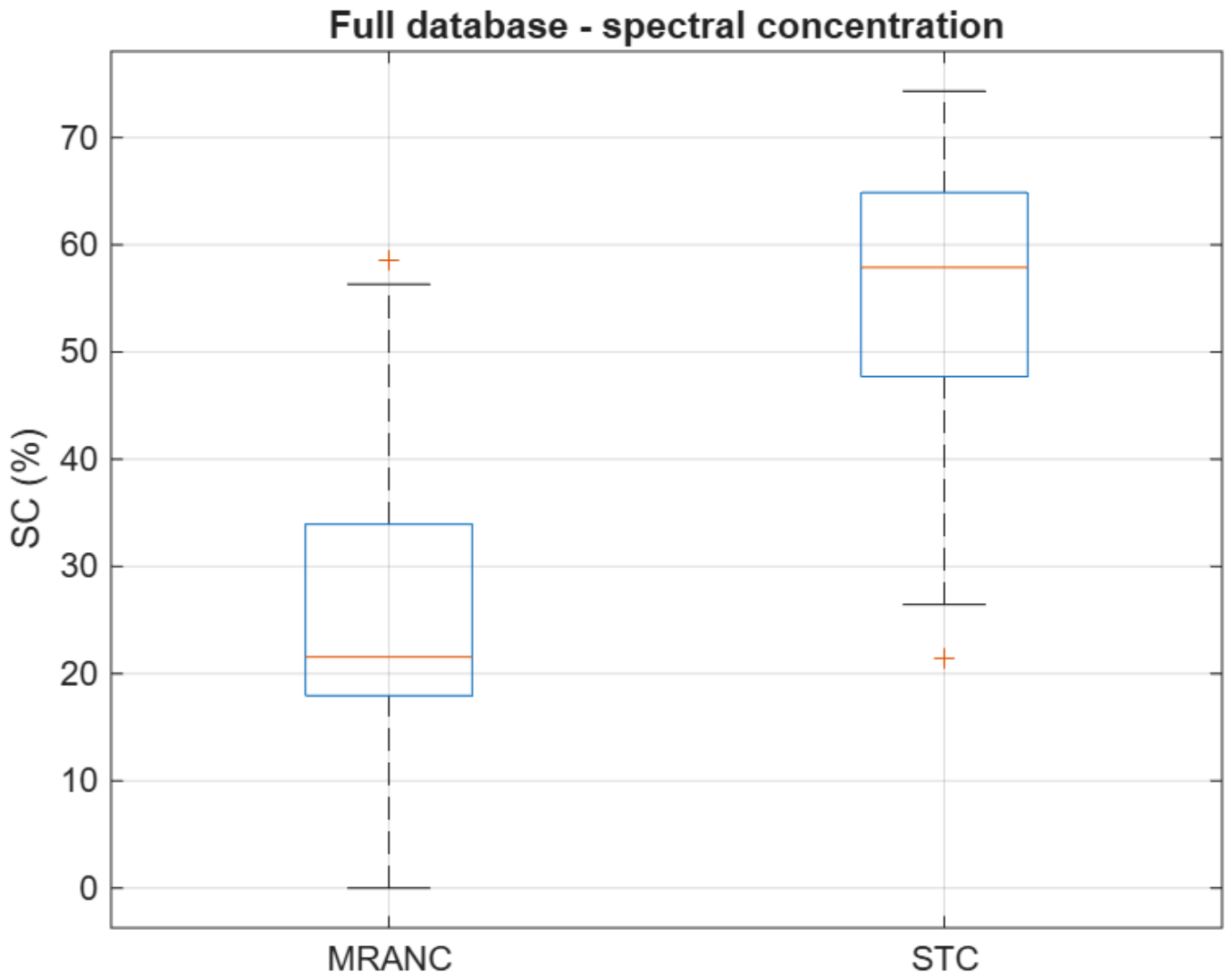


*Figure 18. Full-database comparison of spectral concentration for MRANC and STC. STC is shifted much higher, indicating markedly cleaner atrial estimates.*

**Table 7. Median spectral parameters over labeled patients: MRANC vs STC**

| Method | Median DF (Hz) | Median SC (%) | Implication |
|---|---|---|---|
| MRANC | 5.310 | 21.666 | Captures atrial-related rhythm, but with relatively poor signal organization |
| STC | 6.592 | 58.257 | Produces cleaner atrial estimates concentrated around the AF rhythm |

At the population level, MRANC still produces physiologically plausible DF values in the AF range, which means it is not failing completely. However, its DF values are lower and more spread out than those from STC, and its SC is much lower.

The large SC gap is decisive: STC produces much more organized atrial estimates with less residual ventricular contamination. Therefore, while MRANC is able to retain some atrial-rhythm information, it is clearly inferior to STC for robust AA extraction on this dataset.

# 5. Blind source separation

## 5.1 Synthetic source separation (Assignment 3.1)

The synthetic BSS experiments compared PCA and ICA in two scenarios: (1) when the desired source contribution to the first sensor was reduced to 0.01, and (2) when the desired source leaked into the other observations. In both cases, separation quality was measured using correlation between the recovered component and the true desired source.

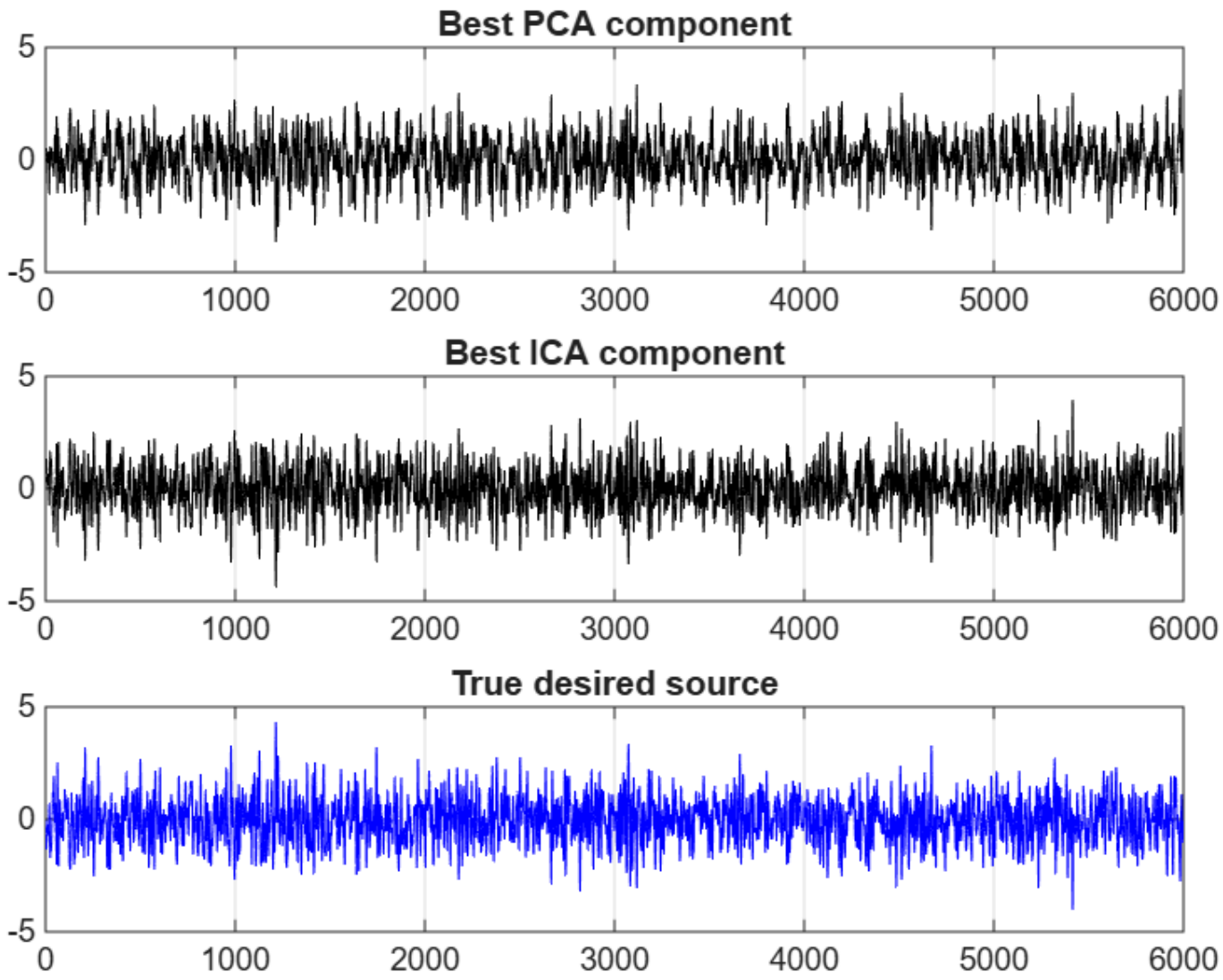


***Figure 19. Synthetic BSS experiment 3.1-1. The best PCA and ICA components are compared with the true desired source when the desired contribution in sensor 1 is reduced to 0.01.***

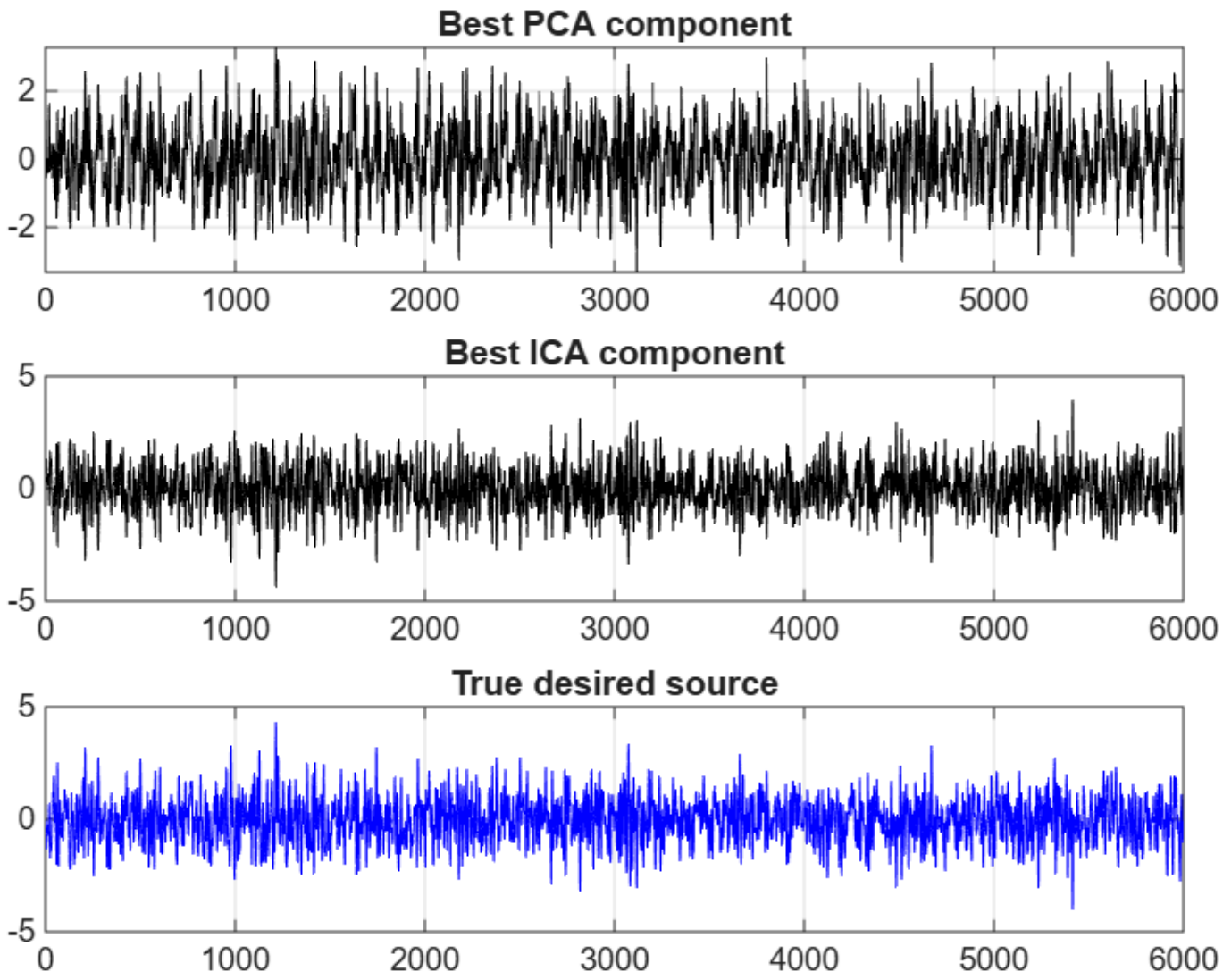


***Figure 20. Synthetic BSS experiment 3.1-2. The best PCA and ICA components are compared with the true desired source when leakage of the desired source is introduced into the other observations.***

**Table 8. Synthetic BSS performance measured by component correlation with the true desired source**

| Scenario | Best PCA correlation (component) | Best ICA correlation (component) | Interpretation |
|---|---|---|---|
| Reduced desired contribution in sensor 1 | 0.793 (component 3) | 1.000 (component 3) | ICA achieves almost perfect recovery; PCA only partially follows the source |
| Leakage into reference-like sensors | 0.733 (component 3) | 1.000 (component 3) | Leakage worsens PCA further, while ICA remains robust in this setup |

These results highlight the conceptual difference between PCA and ICA. PCA is a variance/decorrelation method, so it extracts orthogonal directions of largest energy. When the desired source becomes weak or mixed in a less favorable way, PCA is no longer guaranteed to isolate it cleanly.

ICA, by contrast, aims to recover statistically independent non-Gaussian sources. In both synthetic experiments it achieved a correlation of 1.000 with the true desired source, which indicates essentially perfect recovery up to sign/scale ambiguity. Therefore, ICA is the more appropriate BSS tool for genuine source separation, while PCA should be regarded mainly as a decorrelation/preprocessing method.

## 5.2 Fetal ECG extraction with PCA and ICA (Assignment 3.2)

The complete 8-lead pregnancy ECG was next processed with blind source separation. The goal was to separate maternal and fetal heartbeat sources without using explicit thoracic reference channels. Candidate fetal components were selected from the separated sources based on abdominal/thoracic contribution ratios and plausible fetal heart-rate behavior.

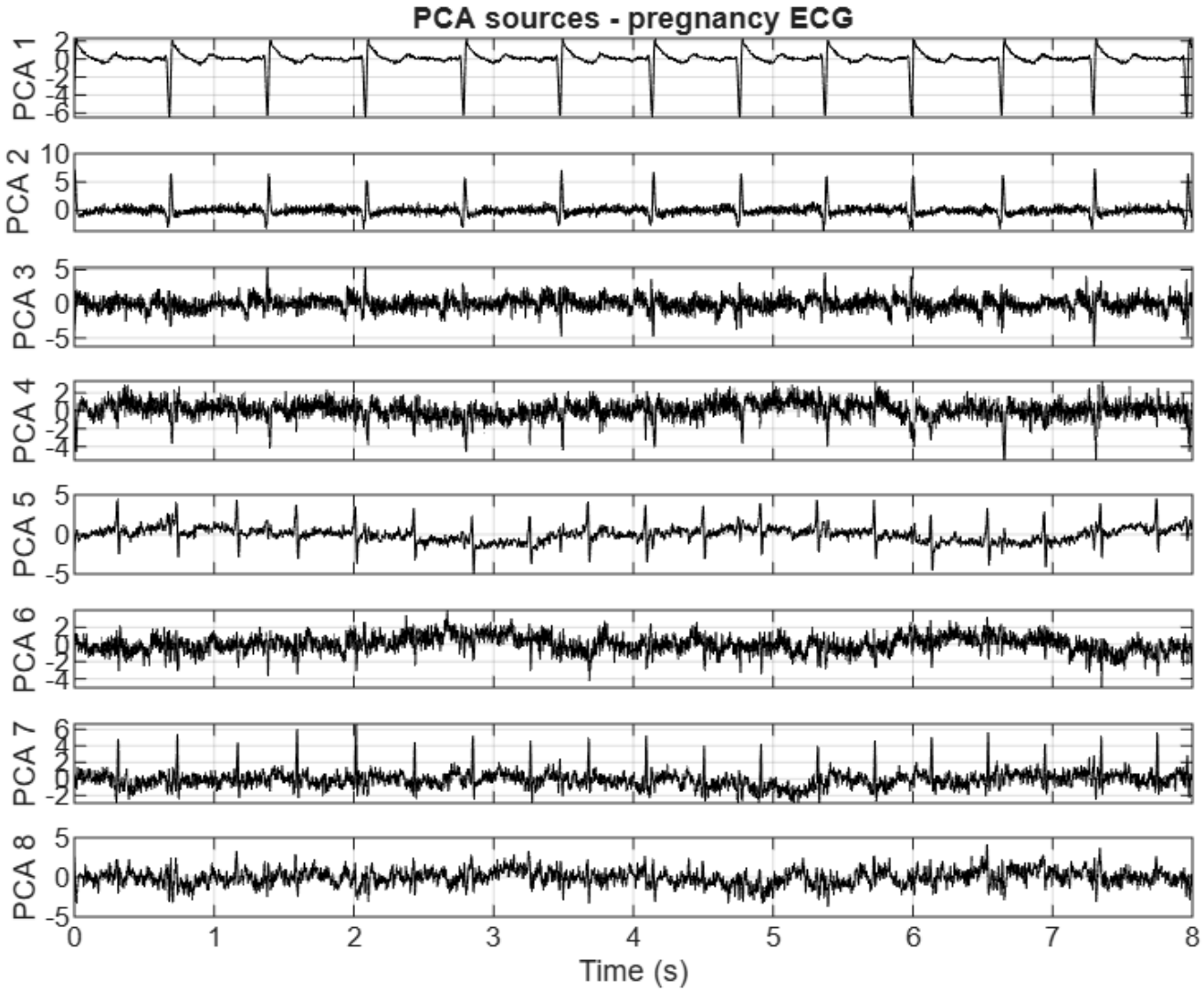


***Figure 21. PCA source signals obtained from the 8-lead pregnancy ECG. Leading components are dominated by stronger maternal activity, while weaker components contain potential fetal information.***

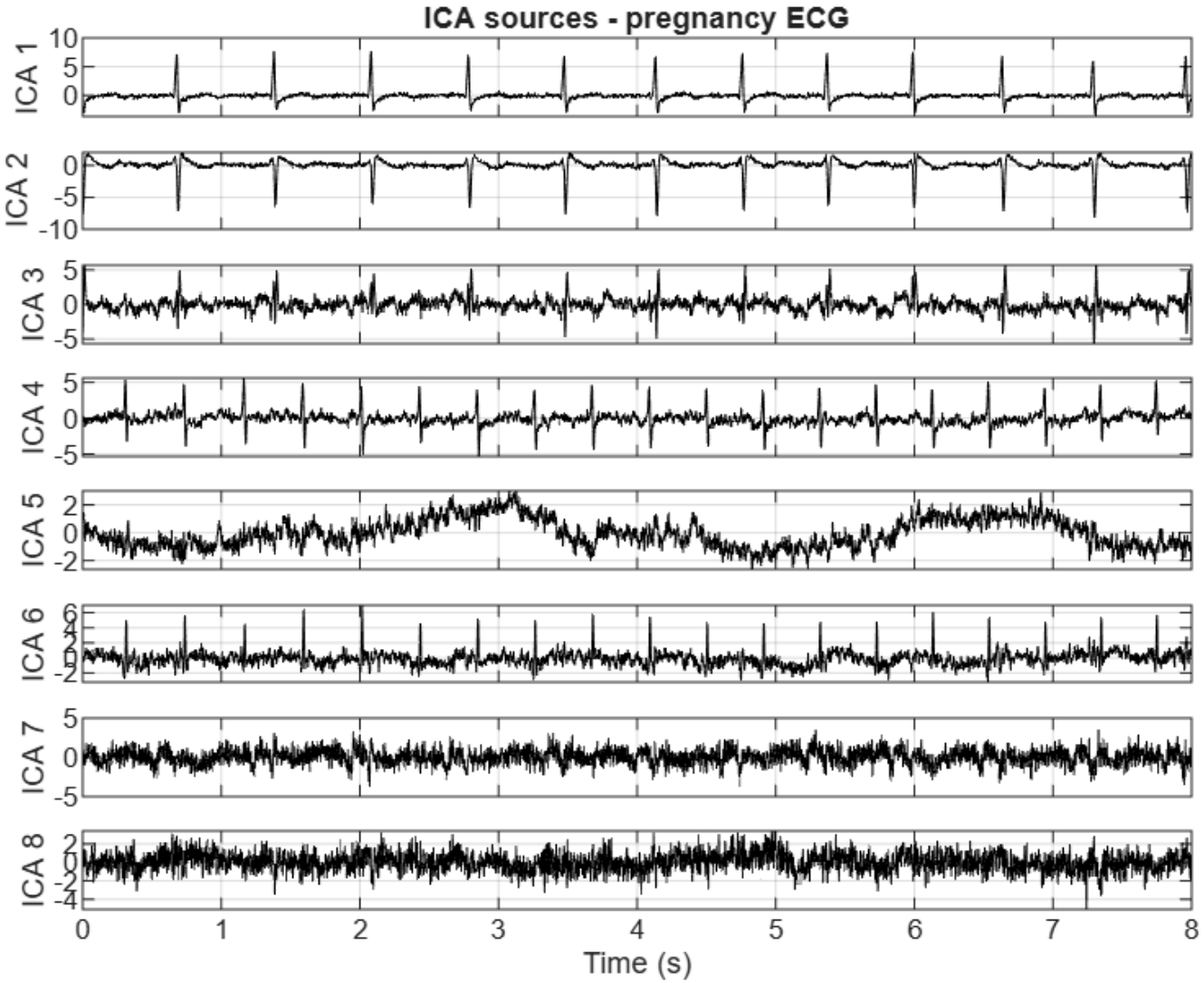


*Figure 22. ICA source signals obtained from the 8-lead pregnancy ECG. ICA produces a more distinct candidate fetal component than PCA.*

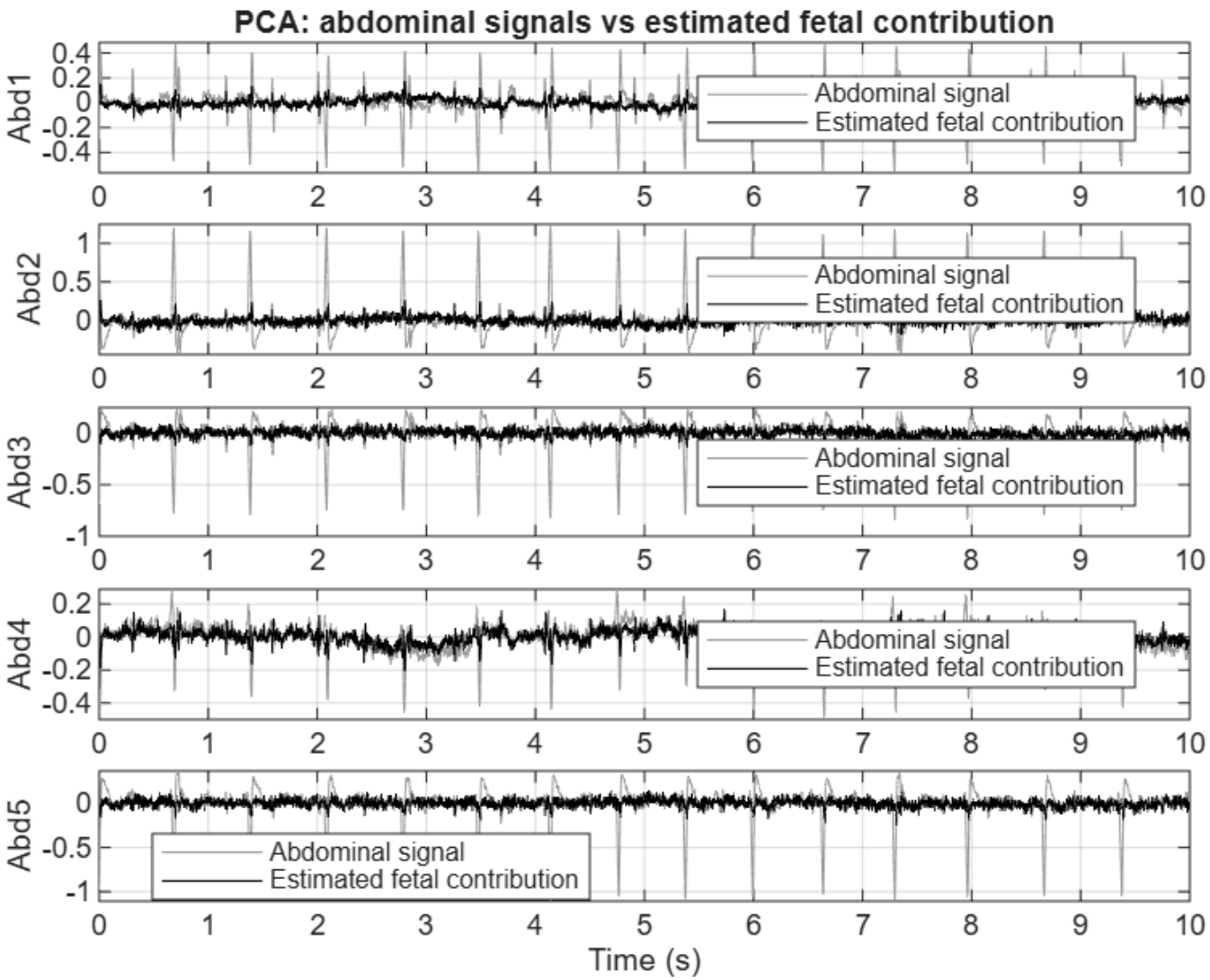


*Figure 23. PCA-based reconstruction of the estimated fetal contribution in the abdominal leads. The contribution is low-amplitude and remains partly mixed.*

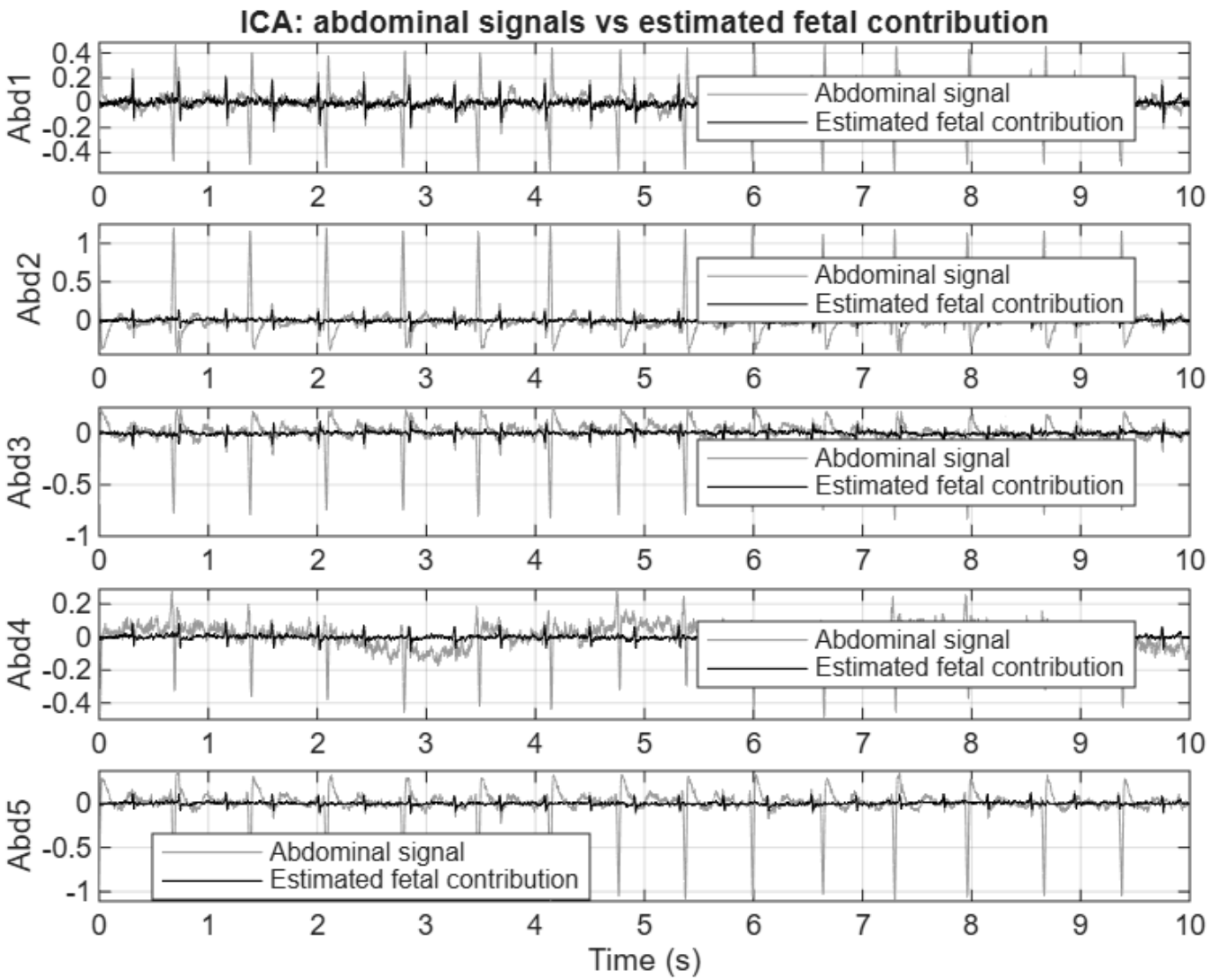


***Figure 24. ICA-based reconstruction of the estimated fetal contribution in the abdominal leads. The extracted component is low-amplitude but appears more distinct than in PCA.***

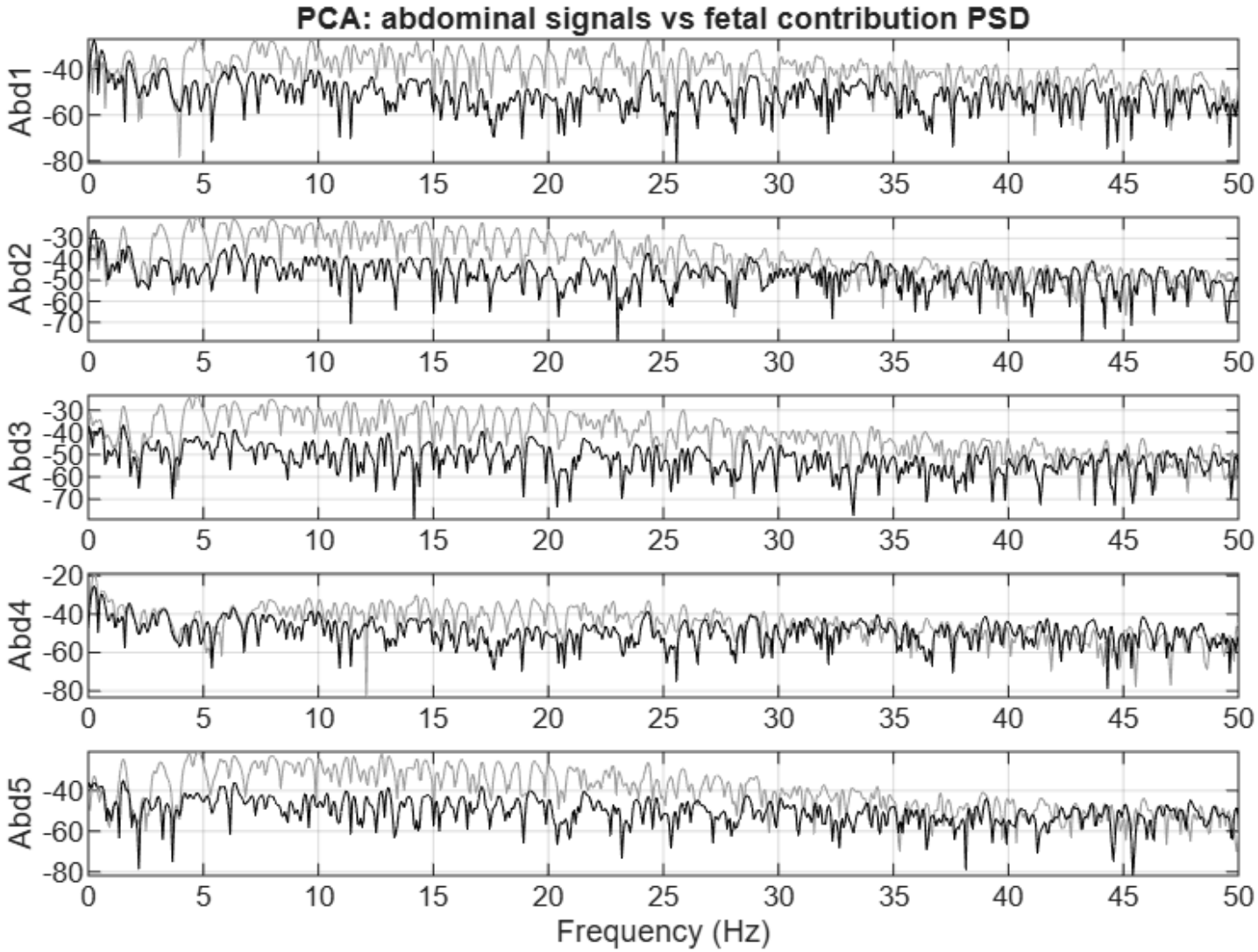


*Figure 25. Welch PSDs comparing abdominal signals and PCA-estimated fetal contributions.*

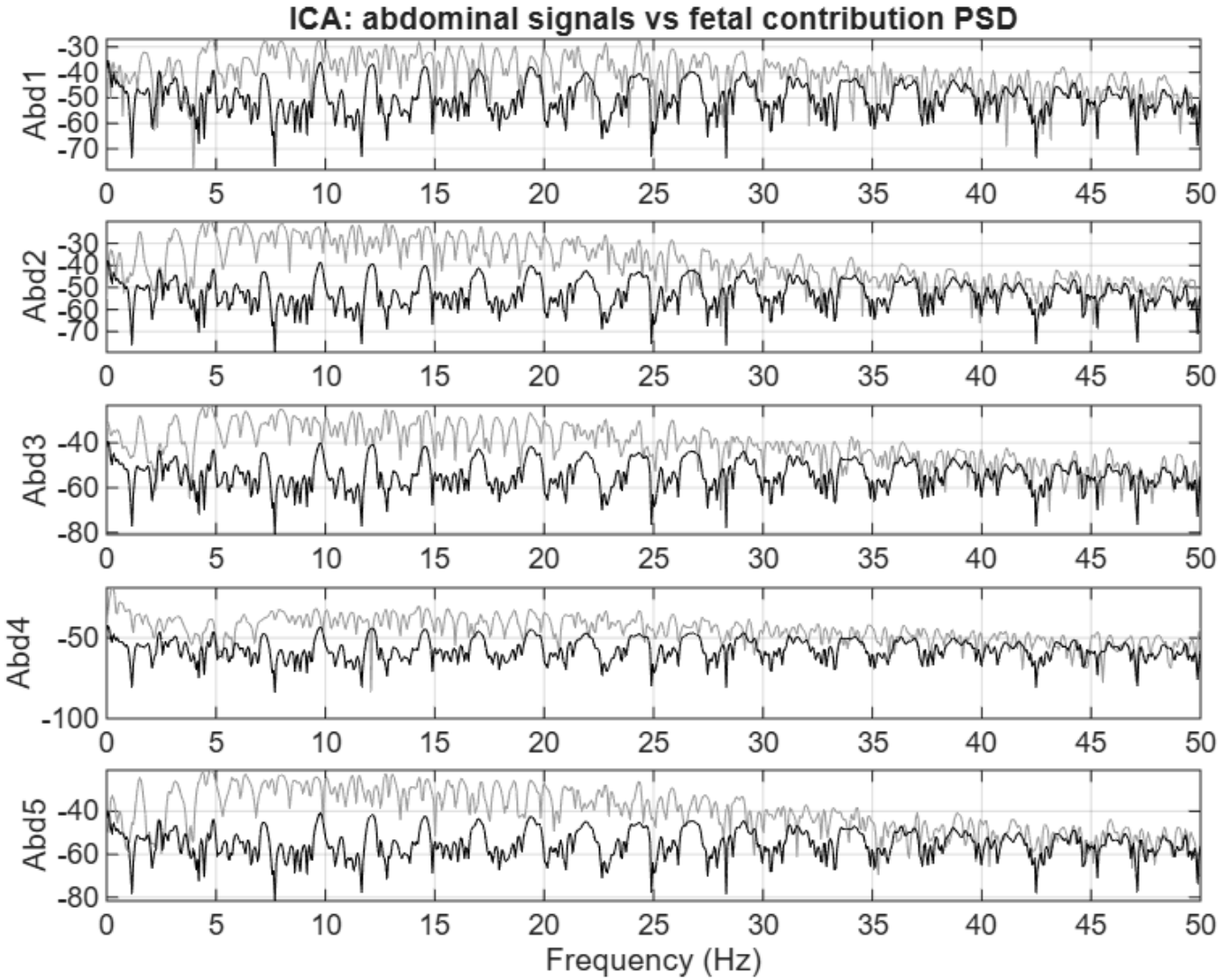


*Figure 26. Welch PSDs comparing abdominal signals and ICA-estimated fetal contributions.*

**Table 9. Suggested fetal components from BSS**

| Method | Suggested fetal component(s) | Interpretation |
|---|---|---|
| PCA | [6, 4] | Fetal information appears spread across weaker orthogonal variance modes |
| ICA | 4 | A single component shows the clearest fetal-like behavior |

The PCA decomposition places the dominant maternal activity in the leading components. The likely fetal information is pushed into weaker components with much lower variance, which makes identification difficult and suggests that PCA does not isolate the fetal source cleanly.

ICA performs better. The suggested ICA candidate is more distinct, and its reconstructed contribution in the abdominal leads appears more structured and less dominated by the large maternal complexes. The PSD comparisons also support this impression: both methods extract low-power components compared with the original abdominal signals, but ICA yields a fetal-related contribution that is cleaner and more interpretable.

Compared with MRANC, BSS has the attractive advantage that it does not require explicit maternal reference channels. In this pregnancy record, ICA offered the best blind extraction among the tested BSS methods, while PCA was weaker and MRANC remained only partially successful. Thus, for fetal extraction in this dataset, the qualitative ranking was ICA > PCA, and ICA also appears more promising than MRANC.

## 5.3 Atrial activity extraction for a single AF patient with PCA and ICA (Assignment 3.3)

The first AF patient was then analyzed with BSS using the six precordial leads. Atrial-source selection was automated by choosing the component with the highest SC among those having a DF inside the AF band [3, 9] Hz.

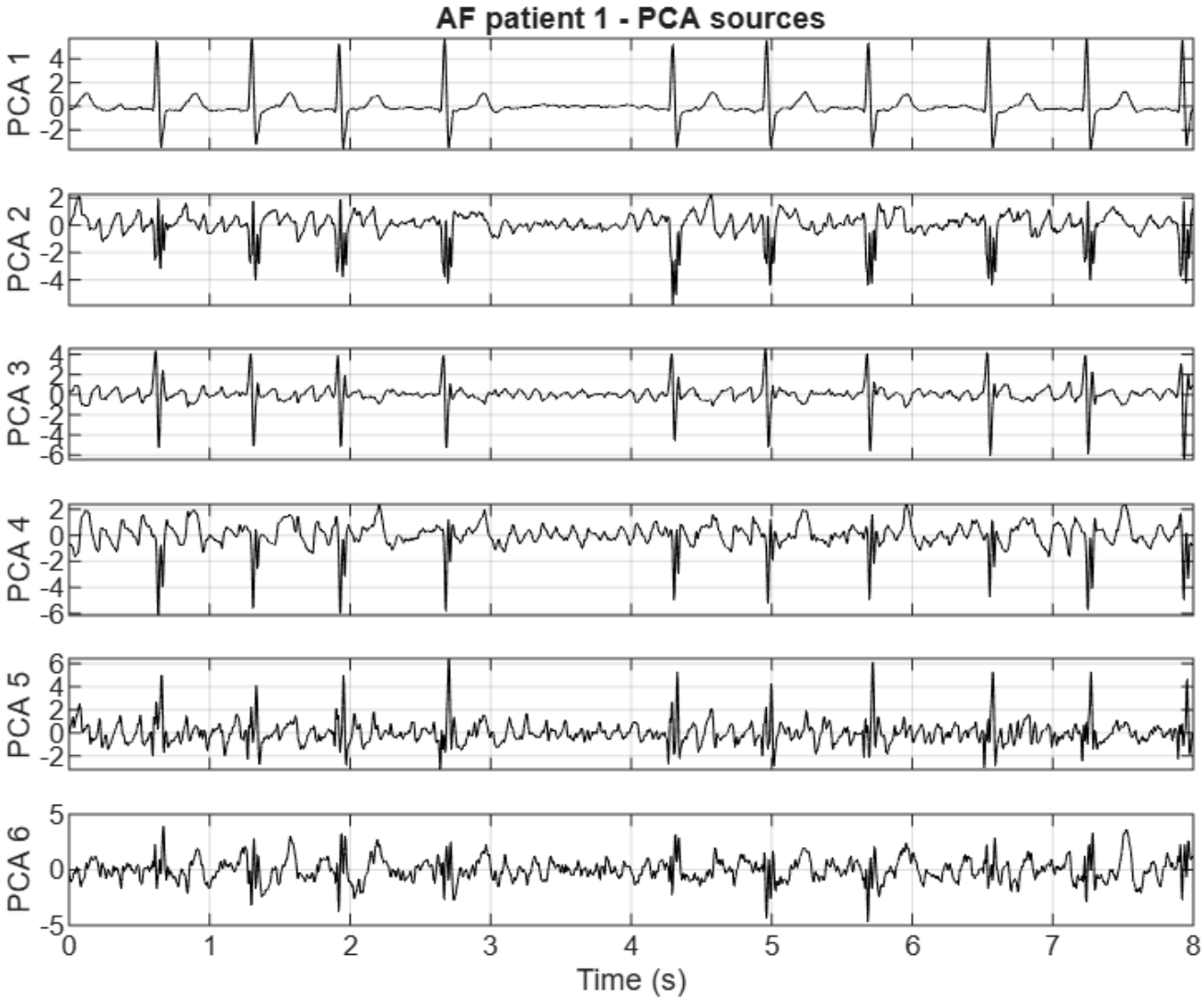


***Figure 27. PCA sources for AF patient 1. Several components still contain clear ventricular morphology, indicating incomplete separation of atrial activity.***

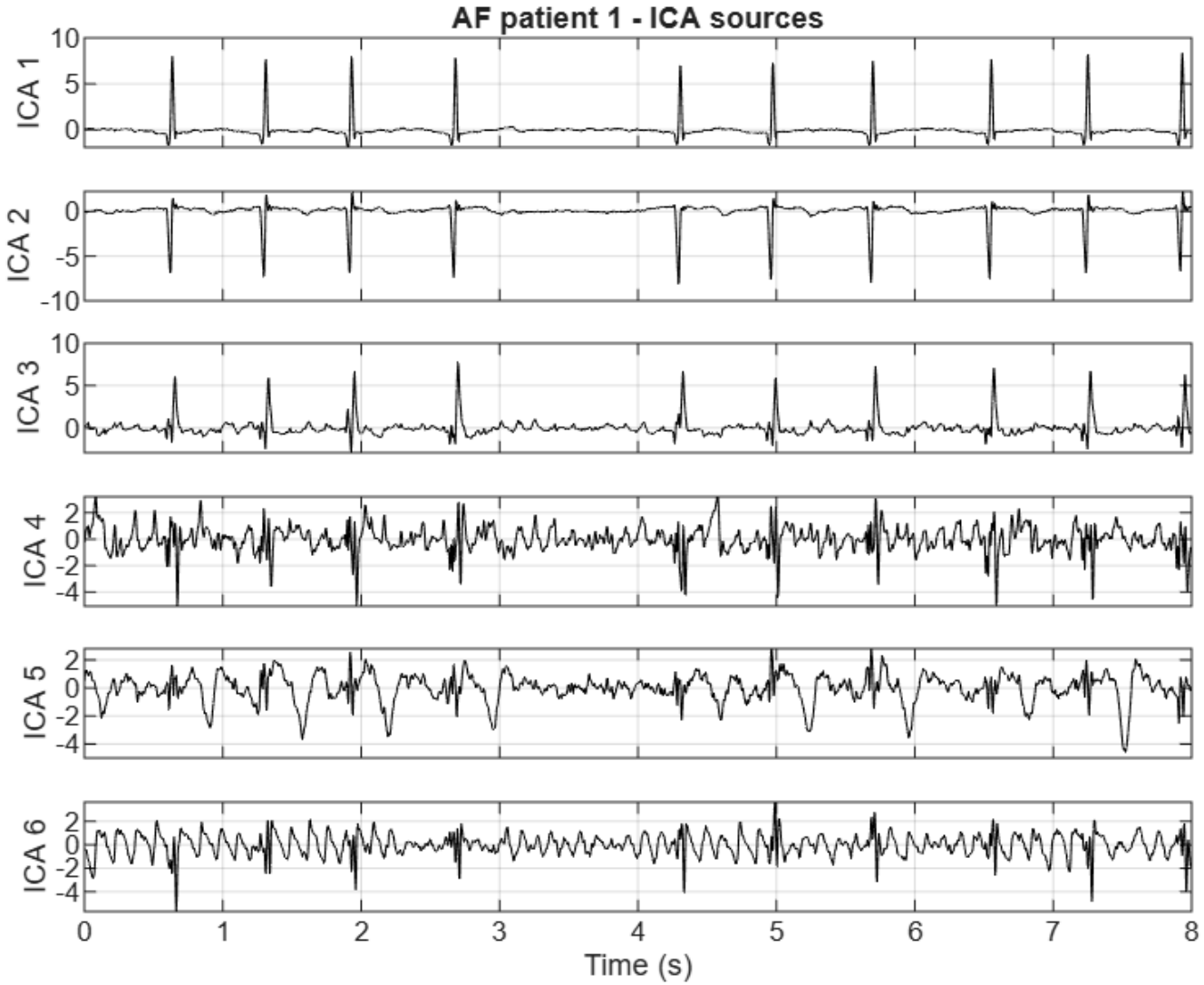


*Figure 28. ICA sources for AF patient 1. ICA produces a more plausible atrial-related component than PCA.*

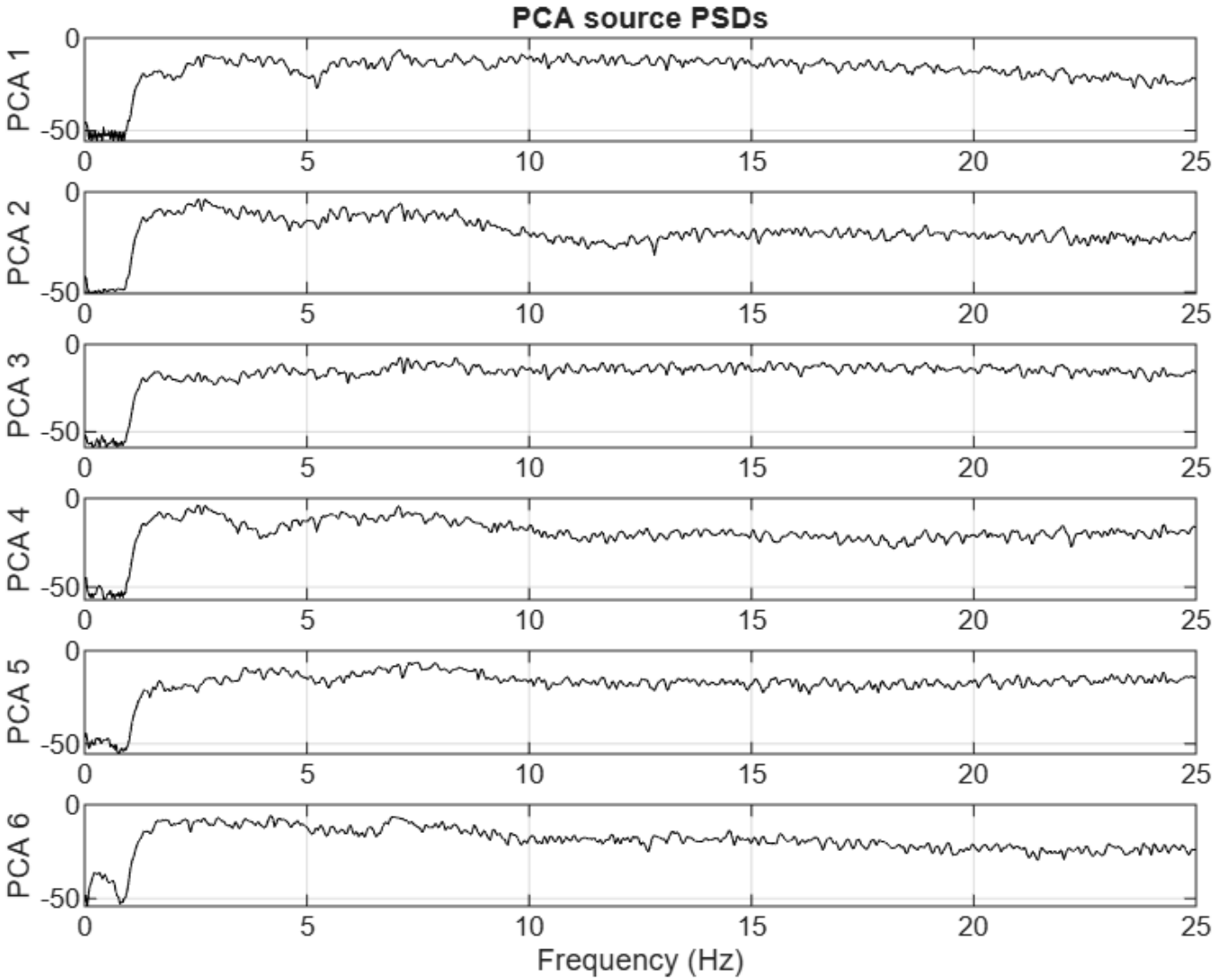


*Figure 29. PSD of PCA source components for AF patient 1.*

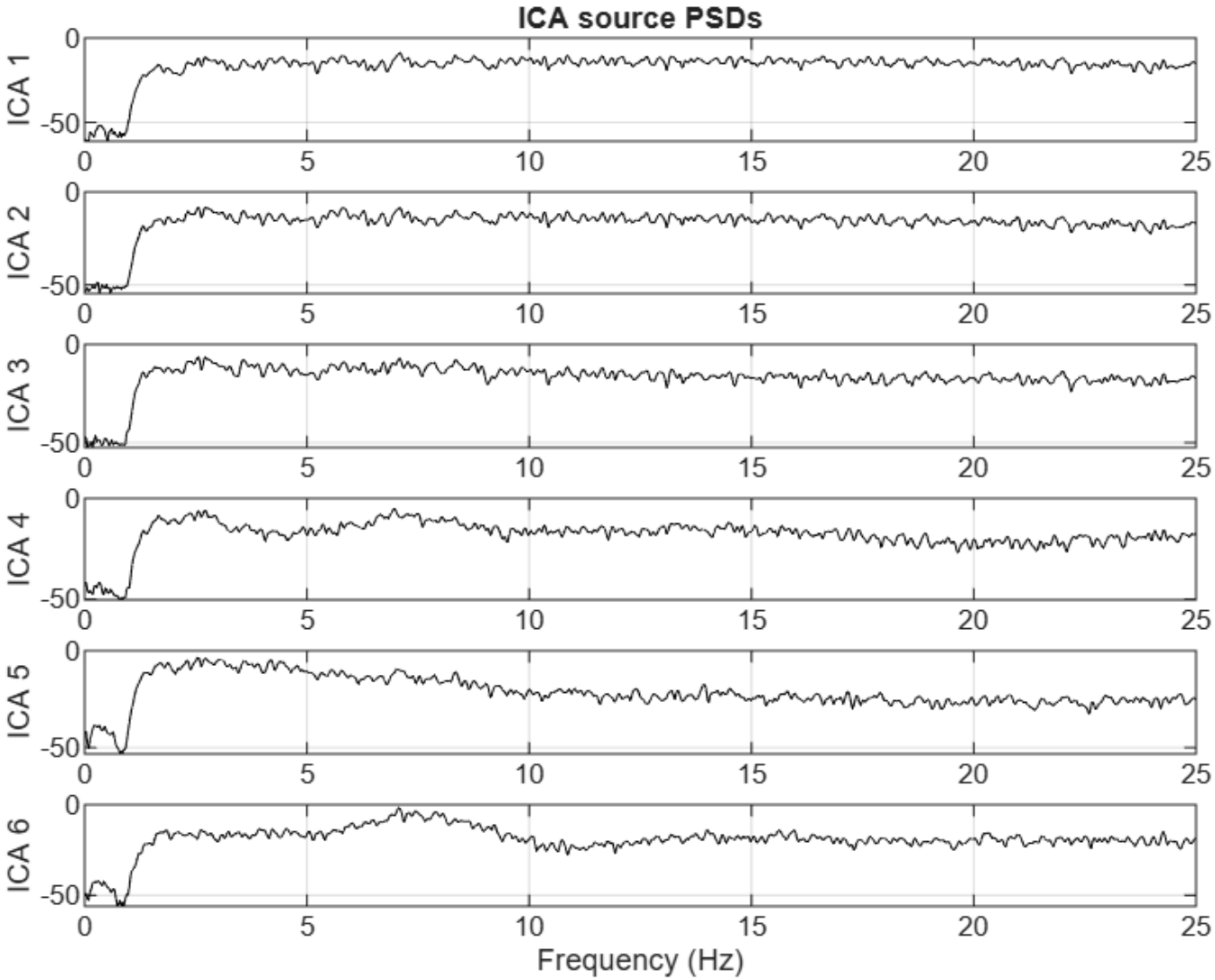


*Figure 30. PSD of ICA source components for AF patient 1.*

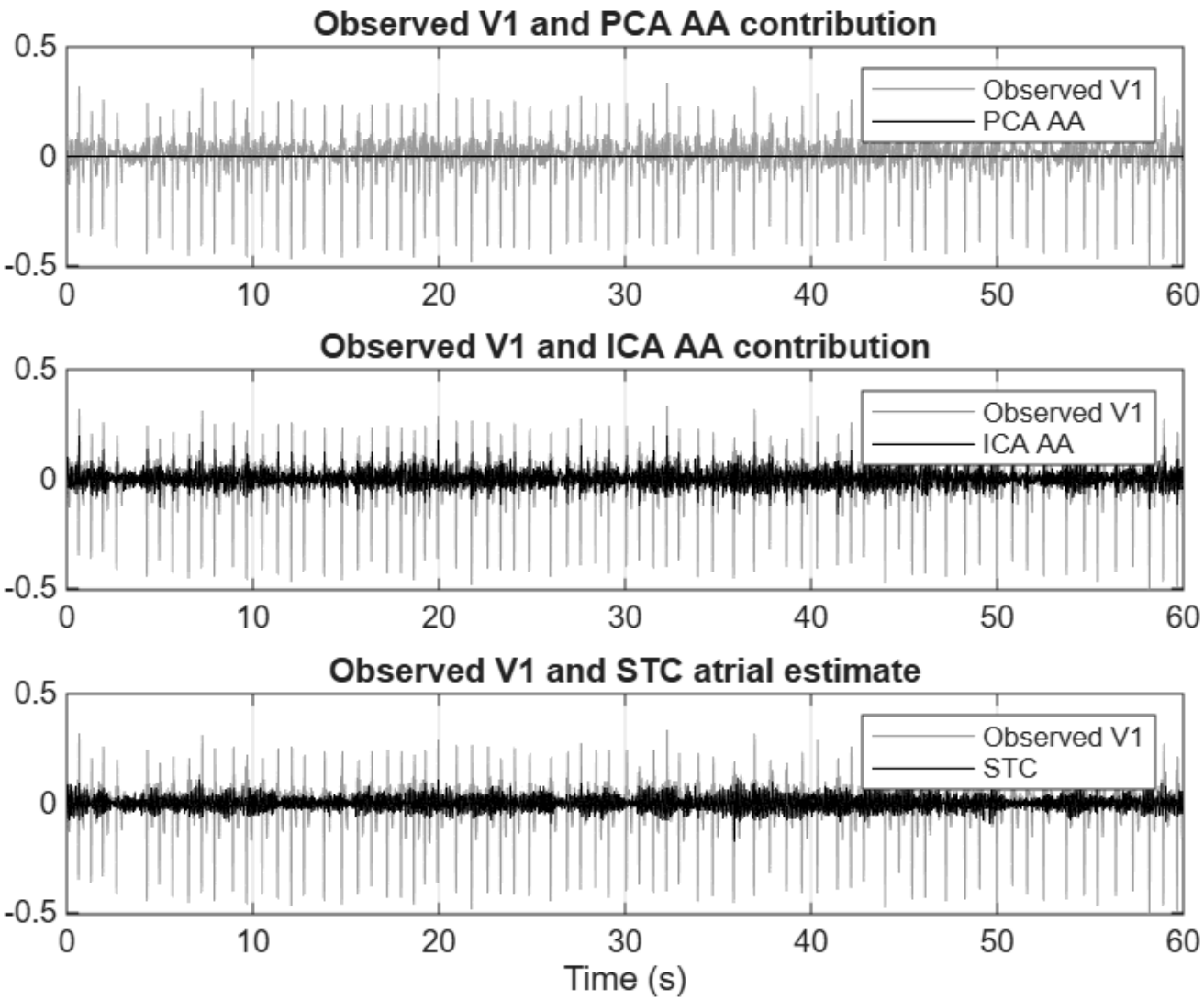


***Figure 31. Observed V1 together with the atrial contribution reconstructed from PCA, ICA, and the STC atrial estimate.***

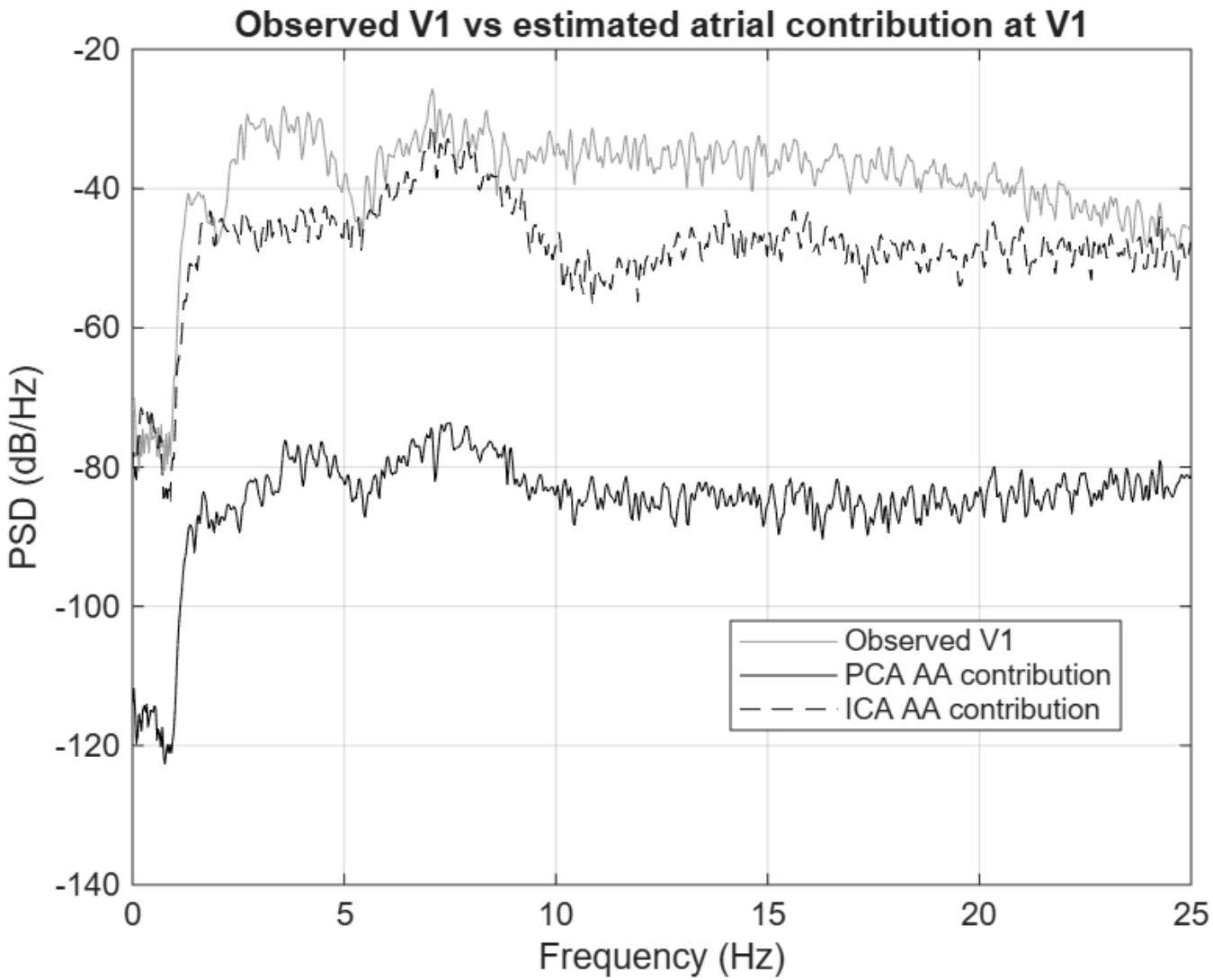


*Figure 32. PSD of observed V1 compared with the estimated atrial contribution at V1 reconstructed from PCA and ICA.*

**Table 10. Automatic atrial-source selection for AF patient 1**

| Method | Selected component | DF (Hz) | SC (%) | Interpretation |
|---|---|---|---|---|
| PCA | 5 | 7.446 | 28.587 | Atrial-related component found, but still weakly organized |
| ICA | 6 | 7.080 | 47.221 | Cleaner and more organized atrial source than PCA |
| MRANC (from Section 4.3) | — | 7.080 | 29.893 | Recovers the main rhythm but leaves contamination |
| STC reference | — | 7.080 | 64.292 | Cleanest atrial estimate among all methods for patient 1 |

Both PCA and ICA identified components with DF in the AF band, so both methods captured atrial-related information. However, the SC values clearly separate their quality. ICA (47.221%) is far more concentrated than PCA (28.587%), meaning the ICA estimate is cleaner and less contaminated.

Comparing all methods for patient 1 yields a clear ranking. STC is best (SC = 64.292%), ICA comes second, and MRANC and PCA are much lower and essentially comparable in purity. All three non-STC methods recover the dominant atrial rhythm reasonably well, but only ICA provides a meaningfully clean blind estimate.

Therefore, for this patient the ranking is STC > ICA > MRANC ≈ PCA. This single-patient result is important because it anticipates the same ordering later observed on the full dataset.

## 5.4 Atrial activity extraction over the full AF database with PCA, ICA, MRANC, and STC (Assignment 3.4)

Finally, the full database was processed and the methods were compared using DF and SC distributions over the 63 labeled patients. This provides the most comprehensive overall performance assessment in the report.

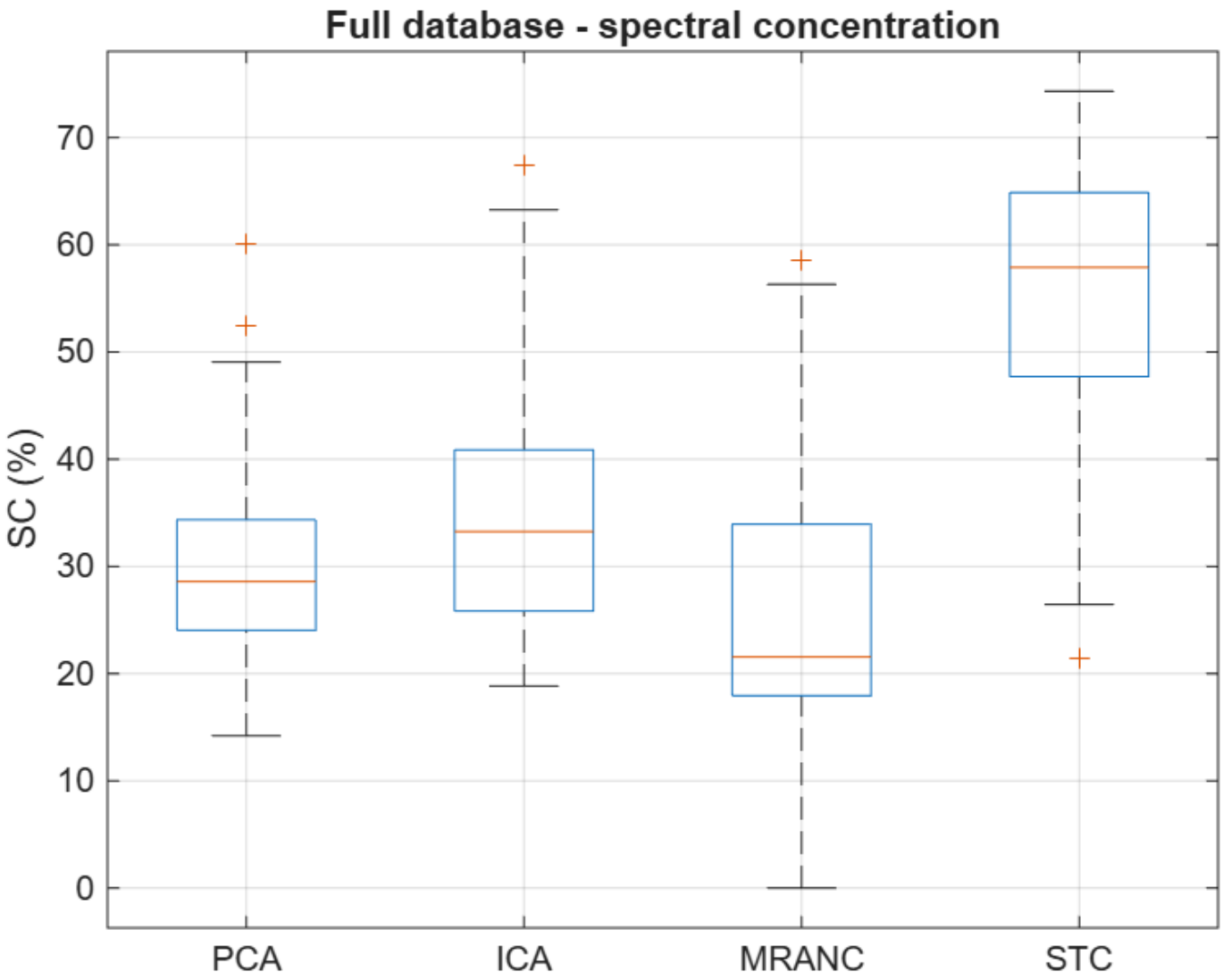


***Figure 33. Full-database comparison of dominant frequency over labeled patients. All methods remain in the physiological AF band, but STC and ICA are the most consistent.***

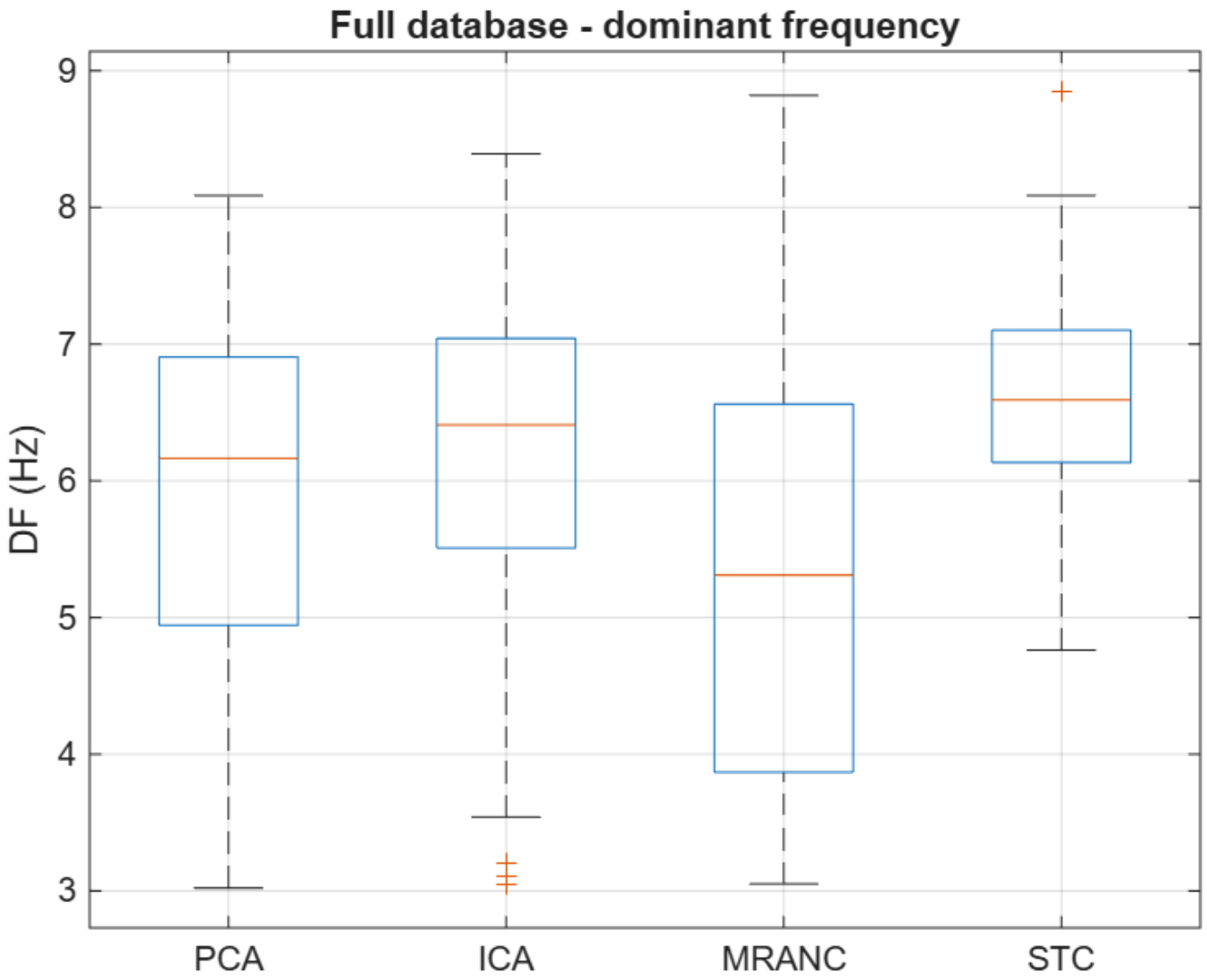


***Figure 34. Full-database comparison of spectral concentration over labeled patients. STC is clearly superior, followed by ICA, then PCA, then MRANC.***

**Table 11. Full-database median spectral parameters over labeled AF patients**

| Method | Median DF (Hz) | Median SC (%) | Overall assessment |
|---|---|---|---|
| PCA | 6.134 | 28.836 | Recovers plausible AF rhythm, but with moderate contamination |
| ICA | 6.409 | 33.442 | Best of the alternative methods; more organized than PCA and MRANC |
| MRANC | 5.310 | 21.666 | Weakest spectral concentration and lower DF |
| STC | 6.592 | 58.257 | Best overall method; cleanest and most organized atrial estimate |

All methods recover median DF values within the AF band, so each one captures some atrial-rhythm information. However, SC reveals the quality differences much more sharply. STC is far superior to all alternatives, confirming that its atrial estimates are the cleanest and most organized.

Among the non-STC methods, ICA is the strongest. Its median DF is close to STC and its median SC is higher than PCA and MRANC. PCA is usable but weaker, while MRANC has the lowest SC and the least stable DF distribution.

The full-database ranking is therefore unambiguous: STC > ICA > PCA > MRANC. This result agrees with the single-patient analysis and with the synthetic BSS experiments, which also showed that ICA is more suitable than PCA for genuine source separation.

## 6. Overall conclusions

The lab assignment produced a coherent set of results across filtering, adaptive cancellation, and blind source separation. The most important findings are summarized below.

- Spectral inspection of the noisy pregnancy abdominal lead Abd4 correctly revealed baseline wander, 50 Hz interference, and broadband high-frequency noise. The chosen filter chain reduced all three contaminants while preserving ECG morphology.
- In AF spectral analysis, the provided atrial estimates Xa were much cleaner than the raw recordings Xva. Xa showed higher DF within the AF band and dramatically higher SC, confirming more organized atrial activity.
- Despite being useful descriptive biomarkers, DF and SC alone were only modest predictors of AF recurrence in the present dataset. The best 5-fold balanced accuracy was 0.555, which is not strong enough for reliable prediction.
- In synthetic ANC experiments, small LMS step sizes were preferable, while too large a step size or an unnecessarily large filter order degraded the output. Leakage of the desired signal into the references was clearly harmful.
- MRANC achieved only partial fetal ECG extraction from the pregnancy record and partial atrial extraction from AF ECGs. It retained useful rhythm information but generally left substantial residual contamination.

- Synthetic BSS experiments demonstrated the superiority of ICA over PCA for actual source separation. ICA recovered the desired synthetic source almost perfectly, whereas PCA only partially followed it.
- For fetal ECG extraction from the pregnancy record, ICA produced the clearest blind estimate among the BSS methods and looked more effective than PCA; both offered an attractive reference-free alternative to MRANC.
- For AF atrial extraction, STC was consistently the best method. Among the tested alternative methods, ICA was clearly superior to PCA and MRANC. On the full AF database, the performance ranking was STC > ICA > PCA > MRANC.

Taken together, the experiments show that no single method is universally optimal for every ECG-processing task. Classical filtering remains indispensable for straightforward denoising; ANC is useful when clean references are available and well matched; and ICA-based BSS is particularly powerful when the goal is to separate latent physiological sources. For the AF dataset used here, the provided STC atrial estimates remain the strongest benchmark, while ICA is the best-performing alternative among the methods tested in this lab.

## 7. Limitations and implementation notes

Two methodological caveats should be explicitly acknowledged. First, the AF sampling frequency was not stored in the uploaded MAT files, so the code used fs = 250 Hz based on the analysis configuration. If the original course dataset uses a different sampling rate, the reported absolute frequency values should be recomputed with that value. Second, the professor's PCA function was used exactly as supplied, but the ICA stage was implemented as a practical ICA-based separation step available in the final working MATLAB package. This does not change the comparative conclusions drawn from the produced figures and metrics, but it should be noted for strict reproducibility.

These limitations do not affect the qualitative conclusions of the lab. The ranking of methods, the interpretation of signal cleanliness via SC, and the observed differences between raw and extracted signals are all strong and consistent across multiple experiments.

## 8. References

Zarzoso, V., & Nandi, A. K. (2001). Noninvasive fetal electrocardiogram extraction: blind separation versus adaptive noise cancellation. IEEE Transactions on Biomedical Engineering, 48(1), 12-18.

Garibaldi, M., & Zarzoso, V. Algorithms for atrial signal extraction in atrial fibrillation ECGs: a comparison based on the correlation between endocardial and surface dominant frequency.

Langley, P., Stridh, M., Rieta, J. J., Millet, J., Sörnmo, L., & Murray, A. (2006). Comparison of atrial signal extraction algorithms in 12-lead ECGs with atrial fibrillation. IEEE Transactions on Biomedical Engineering, 53(2), 343-346.

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